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Review

Federated Learning: Navigating the Landscape of Collaborative Intelligence

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Abstract: As data become increasingly abundant and diverse, their potential to fuel machine learning models is increasingly vast. However, traditional centralized learning approaches, which require aggregating data into a single location, face significant challenges. Privacy concerns, stringent data protection regulations like GDPR, and the high cost of data transmission hinder the feasibility of centralizing sensitive data from disparate sources such as hospitals, financial institutions, and personal devices. Federated Learning addresses these issues by enabling collaborative model training without requiring raw data to leave its origin. This decentralized approach ensures data privacy, reduces transmission costs, and allows organizations to harness the collective intelligence of distributed data while maintaining compliance with ethical and legal standards. This review delves into FL's current applications and its potential to reshape IoT systems into more collaborative, privacy-centric, and flexible frameworks, aiming to enlighten and motivate those navigating the confluence of machine learning and IoT advancements.

Keywords: Federated Learning; internet of things; machine learning; data privacy

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1. Introduction

Federated Learning (FL) is a decentralized machine learning paradigm designed to enable the collaborative training of models across multiple data sources without requiring the transfer of raw data. Unlike traditional centralized approaches, where data from various sources are aggregated into a single server for model training, FL operates by distributing the computation to local devices or nodes. Each participating device trains the model locally using its private dataset and shares only the model updates, such as gradients or weights, with a central server. The central server aggregates these updates to produce a global model, which is then sent back to the devices for further refinement in an iterative process. This mechanism not only preserves data privacy but also reduces the cost and latency associated with transmitting large datasets.

A core principle of FL is its ability to handle Non-Independent and Identically Distributed (Non-IID) data across devices. Unlike centralized systems that assume uniform data distributions, FL operates in environments where data may vary significantly between participants, reflecting real-world conditions such as user-specific preferences or regional trends. Another critical feature is its support for massive-scale deployments, as FL is inherently designed to function across thousands or even millions of devices, such as smartphones or IoT sensors. FL relies on communication-efficient protocols to manage the exchange of model updates, ensuring scalability even with limited network bandwidth. Privacy and security are foundational to FL, with techniques such as secure aggregation, differential privacy, and homomorphic encryption commonly integrated into the framework to safeguard against adversarial attacks and data leakage. Moreover, the adaptability of FL enables it to support various learning paradigms, including supervised, unsupervised, and reinforcement learning, making it versatile across applications [1].

Federated Learning (FL), Traditional Distributed Learning (TDL), and Edge-based Learning (EBL) are distinct paradigms for training machine learning models across multiple systems. FL focuses on privacy-preserving training by ensuring that raw data remain localized on devices while only sharing model updates (e.g., gradients) with a central server for aggregation. In contrast, TDL typically involves partitioning data across nodes and frequently sharing data or gradients with central systems to achieve high-performance and scalable training, often in centralized or semi-centralized setups. EBL, however, prioritizes real-time, low-latency processing by training and performing inference directly on edge devices, such as IoT sensors or smartphones, without significant reliance on centralized servers. While FL and EBL both emphasize keeping data on devices, FL is geared toward collaborative model training across distributed clients, whereas EBL operates independently on resource-constrained edge devices. TDL, on the other hand, is more suited for large-scale training tasks in controlled environments, with less emphasis on privacy or device constraints [2]. Table 1 below highlights the differences between FL, TDL, and EBL.

Table 1. Comparison of Federated Learning with Other Distributed Learning Approaches.

Feature	Federated Learning	Traditional Distributed Learning	Edge-Based Learning
Data Distribution	Data remain on local devices; only model updates are shared.	Data are centralized or distributed across servers.	Training occurs on edge devices; models may or may not aggregate updates.
Privacy	High: Raw data never leaves devices.	Lower: Raw data are often transferred to central servers.	Varies: Raw data may stay local, but privacy is not always a design priority.
Infrastructure	Decentralized, relying on devices like smartphones, IoT, etc.	Centralized clusters with high compute power.	Decentralized; often constrained by device capabilities.
Communication Costs	Periodic sharing of model parameters; designed for low bandwidth.	Significant due to frequent transfer of raw data or gradients.	Lower than Traditional Distributed Learning; varies by aggregation strategy.
Scalability	Highly scalable; supports millions of devices.	Limited by server and network bandwidth.	Limited by the capabilities of edge devices and aggregation methods.
Fault Tolerance	High: Devices can drop out without halting the process.	Moderate: System resilience depends on redundancy in clusters.	Varies: Device failures can significantly impact learning.
Hardware Constraints	Accommodates devices with limited compute power.	Requires high-performance computing environments.	Edge devices may face similar constraints as in FL.

In this paper, we present a comprehensive review of Federated Learning (FL), with a particular focus on its transformative role within Internet of Things (IoT) technologies. Our objective is to provide a clear and in-depth overview of the latest advancements, applications, and challenges in the field of FL, making it accessible and informative for researchers, practitioners, and policymakers alike. We explore a diverse array of studies, ranging from foundational academic research to cutting-edge real-world implementations, offering a holistic view of how FL is shaping the landscape of IoT. This review not only highlights the potential of FL to enhance IoT systems through decentralized and privacy-preserving learning strategies but also examines its implications for scalability, security, and efficiency. By synthesizing recent developments and practical insights, we aim to foster a deeper understanding of FL, inspire further innovation, and support its adoption across various sectors where IoT plays a pivotal role.

To ensure a comprehensive and systematic review of the literature, we adopted a structured methodology for identifying relevant studies on Federated Learning (FL) and

its applications. We consulted reputable scientific databases, including IEEE Xplore, ACM Digital Library, Scopus, and Google Scholar. The search strategy employed combinations of keywords such as Federated Learning, IoT, edge computing, Privacy-Preserving Machine Learning, and Distributed AI. To refine our results, we applied inclusion criteria, selecting studies published in peer-reviewed journals or conferences from the last years, with a specific focus on novel FL techniques, applications, and developments. Exclusion criteria were applied to filter out duplicate studies, non-English articles, and works lacking empirical or theoretical contributions to FL. In total, 191 sources were analyzed, including foundational works and the latest advancements.

2. Federated Learning Issues

The workflow of FL typically begins with the initialization of a global model at the central server, which is then broadcast to the participating clients. Each client uses the global model as a starting point, training it locally on its dataset to capture its unique data patterns. The local training process is governed by optimization algorithms, such as stochastic gradient descent (SGD), which adapt the model parameters based on local data distributions. After training, the clients send their model updates—not raw data—to the server. The server aggregates these updates using techniques such as Federated Averaging (FedAvg), a popular algorithm that combines the client contributions while balancing their varying data volumes and computing capabilities [1]. The updated global model is then redistributed to the clients for the next training round, iterating this process until the model converges or reaches a predefined performance threshold. Effective communication is crucial in the FL workflow, given the need to exchange updates between clients and the server efficiently. To minimize communication overhead, FL employs techniques like compressed updates, where the size of transmitted data is reduced, and client selection, where only a subset of clients actively participate in each training round based on criteria like availability or representativeness of their data. Additionally, robust synchronization protocols ensure that the model updates are aggregated securely and consistently, even in the presence of unreliable or resource-constrained devices.

FL systems are also designed with privacy and security in mind, incorporating methods such as differential privacy to mask individual client contributions and secure aggregation to protect updates during transmission. These features ensure that sensitive information remains confidential throughout the training process. By seamlessly integrating these components into a cohesive workflow, FL enables decentralized learning while addressing the unique challenges posed by distributed data, such as heterogeneity, resource variability, and privacy concerns.

2.1. Blockchain

Blockchain and Federated Learning (FL) can complement each other to enhance security, transparency, and trust in distributed machine learning systems. Federated Learning relies on decentralized data training, while blockchain provides a decentralized, tamper-proof ledger to securely record and manage the processes involved in FL. Blockchain can be used to verify the integrity of model updates, ensuring that only authenticated participants contribute to the learning process.

A noteworthy study [3] focused on a blockchain-based model migration approach specifically designed for IoT devices. Through this, feature extractors were seamlessly transferred from faster devices to their slower counterparts, mitigating the interference from lazy or malicious devices.

Broadening the perspective, a separate investigation introduced the Blockchain-Enabled Secure and Incentive-Based Federated Learning (BESIFL) methodology [4]. Tailored for distributed frameworks such as IoT and mobile edge computing, BESIFL pioneers a decentralized learning environment. Its foundation encompasses strategies for malicious node detection, adept reward management, and the selection of reliable nodes. Preliminary results underscored the significant advantages BESIFL offers over traditional systems.

Shifting the lens to the healthcare sector, researchers have unveiled a platform merging Federated Learning with private blockchain technology [5]. This platform aims to bolster data privacy and network adaptability, particularly in wearable IoT devices. Initial evaluations highlighted the platform's prowess in safeguarding users' privacy while upholding the integrity of predictive services.

In the realm of cybersecurity, a captivating study offered a novel approach to identify and leverage application vulnerabilities [6]. This method empowers potential attackers with tools to infiltrate system vulnerabilities, achieved through a composite of communication and analysis techniques. Diving into specialized applications, the BLADE initiative stands out [7]. This blockchain-empowered Federated Learning model specifically targets blade icing detection challenges. Beyond enhancing privacy, BLADE employs a unique imbalanced learning algorithm to rectify class imbalances in sensor data, showcasing superior performance metrics when pitted against conventional methods.

On parallel lines, an adaptive framework integrating Federated Learning and blockchain has been conceptualized [8]. Its primary aim is to solidify trustworthiness and bolster security across networks. Simulation-driven evaluations commended its high accuracy, boasting impressive detection rates. A groundbreaking study zeroed in on data privacy concerns in IIoT, particularly in the context of failure detection [9]. By harnessing a blockchain-based Federated Learning mechanism, the research showcased satisfactory performance metrics via a meticulously crafted prototype.

In a bid to counter potential manipulations of local models within IIoT applications, a distinct framework was formulated [10]. It combines blockchain technology with SGX-enabled processors, with evaluations reaffirming its promise, especially when working with CNN models. Merging the fields of social media and IoT, Differentially Privacy Blockchain-based Explainable Federated Learning (DP-BFL) has been proposed [11]. Positioned as a decentralized network, it preserves user privacy while simultaneously elevating accuracy standards. Its resilience, even in the presence of malicious entities, is proof of its robustness and effectiveness.

Furthermore, exploration led to the proposal of the Blockchain-Enhanced Federated Learning Market [12]. Designed to streamline data sharing and optimize training among IoT devices, it artfully incorporates encrypted model training with differential privacy, all underpinned by a contribution-driven Delegated Proof of Stake consensus mechanism. The system's holistic design has yielded remarkable performance enhancements. From a security standpoint, the Chameleon Hash-Based Variable Trapdoor (CHCT) framework offers a fortified Federated Learning environment for IIoT [13]. Supported by detailed security analyses, the CHCT system's efficacy was exemplified using a redactable medical blockchain.

Incorporating 5G technology, a novel B5G-backed model catered to the Internet of Sensor Things has emerged [14]. By leveraging Federated Learning combined with SVM and RF classifiers, the model excels in malicious node detection as well as feature evaluation and cascading techniques to ensure secure service provisioning. Further evaluations corroborated the superior performance of the RF classifier across various parameters.

Last but not least, researchers introduced Cross-cluster FL (CFL), a Federated Learning paradigm facilitated by cross-chain technology [15]. With a focus on optimizing system productivity, CFL reconceptualizes large BFL clusters into interconnected, smaller units via a cross-chain consensus protocol. The viability of CFL has been authenticated through rigorous experimentation.

2.2. Robustness Against Poisoning

A significant privacy concern arises from the transmission of model updates (gradients or weights) between clients and the central server. These updates, though seemingly abstract, can leak sensitive information about the underlying data through techniques such as gradient inversion or model reconstruction attacks, where adversaries attempt to infer client data by analyzing the updates. This threat is particularly concerning in applications

like healthcare or finance, where even minimal data leakage can have severe consequences. Adversarial attacks in FL can also compromise the learning process itself. Model poisoning attacks involve malicious clients submitting corrupted updates to manipulate the global model, potentially degrading its performance or embedding backdoors that enable adversarial control during deployment. These attacks can be subtle, bypassing conventional aggregation methods to remain undetected while achieving their objectives. Similarly, data poisoning attacks corrupt the local training data of participating clients to bias the global model toward incorrect predictions or specific adversarial goals. To address these challenges, FL incorporates several defense mechanisms. Secure aggregation protocols ensure that updates are encrypted during transmission and can only be decrypted after aggregation, preventing the server from accessing individual client updates.

In a breakthrough for secure Federated Learning, researchers have put forth AutoGM, a secure aggregation rule, and its two related secure FL solutions: AutoGM_FL and AutoGM_PFL [16]. These innovations have demonstrated considerable resilience against model and data poisoning attacks, signaling a new level of robustness in FL systems. Meanwhile, the DisBezant mechanism has emerged as a cutting-edge solution for ensuring Byzantine-resistant Federated Learning in IoT-enabled Multi-Tier Systems (MTS) [17]. Employing a credibility-based framework and designing an efficient privacy-preserving gradient aggregation protocol, DisBezant has proven its ability to effectively defend against Byzantine attacks, as confirmed by several experiments.

The novel RobustFL method has showcased its potency by defending against poisoning attacks on three benchmark datasets [18]. The success of these defenses has proven RobustFL's status as an effective and efficient approach to secure Federated Learning. A significant milestone has been reached with the introduction of an FL-oriented dataset tailored for SSDF attack detection – a first in the field [19]. Investigations into the efficacy of FL in monitoring SSDF incorporated various sensors, attack types, FL scenarios, adversary percentages, and aggregation functions.

Despite FL's promising role in detecting SSDF, it was found to be susceptible to adversaries beyond a 16% threshold. Moreover, binary classifiers were compromised with adversary influence over 33%, while the coordinate-wise median scheme excelled in anomaly detection. Addressing the security vulnerabilities within the Industrial Internet of Things, a multi-tentacle Federated Learning (MTFL) framework has been introduced [20]. It encompasses the tentacle distribution-based efficient poisoning attack detection (TD-EPAD) algorithm and a stochastic tentacle data exchanging (STDE) protocol, mitigating the impact of malicious data. Additionally, the integration of differential privacy technology further strengthens data protection. Prototypes have evidenced a 40% improvement in the global model's accuracy during attack simulations.

In the field of cyberthreat hunting for IIoT environments, an ensemble-based deep Federated Learning model has been proposed [21]. This model refines threat detection by identifying risks via Federated Learning, obviating the need for centralized data exposure during training. Tested against two scenarios, this model not only achieved a higher F1-score relative to existing models but also maintained stability irrespective of the number of connected devices, along with boasting a quicker training time.

A novel framework, BoEI, harnesses the synergy of decentralized Federated Learning (Fed-Trust), blockchain technology, and fog computing to detect cyberattacks within IIoT [22]. It operates on a temporal convolutional generative network paired with semi-labeled data to identify malicious activities. Validated by two public datasets, BoEI's robustness and effectiveness are commendable. In addition, an innovative intrusion detection system known as FELIDS has been tailored for Agricultural IoT infrastructures [23]. It leverages the power of deep learning classifiers, achieving high accuracy in anomaly detection, as evidenced by experiments conducted on public datasets.

Investigations into the role of data augmentation for enhancing anomaly detection in IoT have yielded promising outcomes [24]. By experimenting with a variety of sampling methods to support Machine Learning algorithms, a significant improvement in the ac-

curacy of anomaly detection is highlighted, thereby offering a robust strategy for better safeguarding IoT ecosystems.

In the pursuit of advancing IoT security, a novel and decentralized Federated Learning framework has been conceptualized, ensuring the privacy of IoT data while enhancing anomaly detection [25]. Through simulations employing real-world datasets, this approach has outperformed conventional methods in terms of robustness, accuracy, and efficiency, marking a substantial step forward in IoT security measures. Delving into the challenges of intrusion detection within IoT networks, researchers have put forth a Federated Learning-based method that upholds the privacy of user data [26].

The integration of an ensembler component, which aggregates updates from various entities, has been proven to not only bolster the global model's accuracy but also to ensure superior protection of data privacy when compared with traditional machine learning techniques. An advanced deep learning structure, predicated on Federated Learning, has been tailored for swift and precise anomaly detection within the Industrial Internet of Things (IIoT) [27]. The framework's innovative design incorporates an attention mechanism-based Convolutional Neural Network fused with a Long Short-Term Memory (AMCNN-LSTM) model, which excels in identifying faults. Furthermore, the adoption of a top-k selection-based gradient compression scheme has substantially heightened communication efficacy. Rigorous testing confirms that this framework stands as a highly efficient solution for fault detection in IIoT applications.

Exploring the frontiers of IoT security, a new study has revealed a Federated Learning approach for malware detection that preserves data privacy without sacrificing performance [28]. When pitted against traditional centralized methods, the federated approach demonstrated a comparable efficacy in detecting malware, thus marking a significant stride in safeguarding security without compromising the integrity of data.

Delving into the realm of the Internet-of-Medical-Things (IoMT), researchers have put forth an innovative distributed learning technique that anticipates the presence of a malicious jammer [29]. By employing a sophisticated Stackelberg game and tapping into Difference-of-Convex programming alongside dual decomposition, the team successfully delineated a joint best response strategy. The efficacy of this algorithm was validated through comprehensive simulations, showcasing its potential in a challenging IoMT environment. A breakthrough has been made with the development of the D2MIF mechanism, which ingeniously integrates an isolation forest (iForest) with reinforcement learning (RL) to identify and neutralize malicious models in AIoT applications [30]. Rigorous testing with Mnist and Fashion_Mnist datasets confirmed the mechanism's capability to effectively sieve out harmful models, thus protecting the accuracy of global models in Federated Learning frameworks. In the landscape of IoT, an advanced intrusion detection scheme has been introduced, underpinned by Federated Learning [31]. This scheme fortifies security by confining training and model inferencing to local nodes while enabling the exchange of enhanced models with remote servers.

2.3. Protocols

The intersection of Federated Learning, edge computing, and IoT has led to the development of innovative FL protocols tailored for modern and forthcoming IoT networks. One study in particular emphasized the benefits of these protocols through rigorous experimentation, specifically evaluating accuracy, time, and latency metrics [32]. The field of heterogeneous IoT systems saw the introduction of a specialized FL policy [33]. By incorporating ISODATA for client scheduling and MILP for resource block allocation, this strategy demonstrated a substantial 55.22% increase in accuracy and an impressive 3.62× enhancement in processing speed.

Moreover, Federated Learning's synergy with fog/edge computing has been explored to target vulnerabilities of robotic networks in [34]. The devised strategy identified compromised robot networks primed for back-end web attacks. This method showcased a surge in detection accuracy by 47%, a significant 72% reduction in mitigation time, albeit at a

cost that was 2.7 times higher. Addressing concerns around communication frameworks in FL, researchers unveiled a novel super-wireless-over-the-air FL platform rooted in 6G technology [35]. This innovative solution offers private data training via interference-immune radio waves, thus refining AI performance in the IoT sector.

For the Industrial Internet of Things (IIoT), a unique quantization-based distributed Federated Learning (Q-DFL) approach was introduced [36]. This two-phased technique, comprising local training and synchronization, was instrumental in training a shared machine learning model. Through simulation, it was evident that this method circumvented the limitations of centralized FL and reduced data transmission volumes.

An innovative leap in IoT security came with the unveiling of FedTor, an anonymous Federated Learning framework [37]. Capitalizing on the Tor network's strengths, this solution proposed a streamlined shared key creation mechanism and a reputation-centric router selection methodology. Its efficacy in reducing malicious router occurrences was validated through simulation.

The transformative potential of Federated Learning in distributed systems was spotlighted in a recent article [38]. This piece proposed a serverless learning approach merged with an FL framework, emphasizing the power of consensus-based strategies and local data operations for enhanced inter-device collaboration. In an era demanding robust AI models and fair market practices, a novel algorithm has been proposed [39]. This Iterative Double Auction-based algorithm, fortified with experience-weighted attraction learning, showcased its supremacy over traditional algorithms through simulation, forecasting significant enhancements in AI model quality and market fairness.

Lastly, addressing the challenges faced by IoT devices with limited computational resources and inconsistent communication networks, researchers formulated an asynchronous Federated Learning model [40]. Comprehensive experiments highlighted this model's superiority, especially in diverse settings with varying data distributions.

In recent research, a Privacy-Preserving and Traceable Federated Learning (PPTFL) framework was introduced, incorporating Federated Learning, blockchain, and IPFS [41]. This innovative approach aimed to mitigate the risks of data leakage, model tampering, and algorithmic overhead within the Industrial Internet of Things (IIoT) frameworks. Through rigorous testing, PPTFL demonstrated its effectiveness in addressing these critical concerns.

Exploring further into privacy preservation, a novel scheme tailored for IoT devices emerged [42]. This scheme, characterized by its lightweight nature, was carefully designed to operate within the constraints of limited computational and communication resources typical of IoT devices. It succeeded in safeguarding local data privacy, with its security integrity confirmed through testing on actual IoT devices. Delving into the world of IoT perception data, two advanced Federated Learning models, FedProLs and ProLs, were developed [43]. These models prioritized privacy while efficiently predicting IoT perception data. Upon evaluation against pre-existing prediction models using real-life IoT datasets, both FedProLs and ProLs surpassed their counterparts in performance.

The study of differential privacy techniques revealed their potential in crafting an intrusion detection system based on Federated Learning for industrial IoT networks [44]. The novel Fed+ approach, alongside the traditional FedAvg, underwent comparison on the ToN_IoT dataset. Despite the deliberate introduction of noise into the federated training process, Fed+ maintained comparable accuracy, showcasing its robustness. A solution known as PCFL has been put forward to confront the dual challenges of network bandwidth and privacy attacks inherent in Federated Learning systems within IoT environments [45]. PCFL utilized gradient spatial sparsification, bidirectional compression, and a privacy-preserving protocol. This combination aimed to safeguard data while optimizing for high efficiency in communication. Attention was turned to a multi-party, privacy-centric architecture for the IIoT, adept on fostering collaboration [46]. An edge-enabled IIoT case study was instrumental in assessing the efficacy of this model, revealing how a multi-party platform adhering to FAIR principles could yield significant advantages.

The intersection of Federated Learning and local differential privacy (LDP) was explored to counter threats and communication costs brought on by the growing numbers of Internet of Vehicles (IoV) units [47]. Here, four LDP mechanisms were considered, leading to the creation of a hybrid approach. This novel mechanism coupled with an LDP-FedSGD algorithm enabled a coordinated effort between cloud servers and vehicles for model training. Experimentation affirmed that these algorithms effectively preserved privacy while assuring data utility. As IoT, Information-Centric Networking (ICN), and Federated Learning (FL) converged, they promised enhanced security in human–machine interactions [48]. However, the full implications of their combined application required further investigation to be fully understood.

A cutting-edge hierarchical IoT network model named μDFL was put forth, running on microchain technology to bolster security within Federated Learning landscapes [49]. This system, leveraging an interchain network, sought to ensure both accuracy and privacy and demonstrated scalability and reliability. Through trials, the feasibility of μDFL was validated in various FL device configurations . In the realm of industrial IoT, Verifiable Federated Learning (VFL) emerged as a novel method for big data analysis while preserving privacy [50]. VFL utilized interpolation points for gradient accuracy verification and obfuscation to limit data leakage. Its static verification overhead remained consistent, independent of participant numbers, and evaluations affirmed its reliability and effectiveness.

A recent proposal introduced an asynchronous grouped Federated Learning framework (PAG-FL), addressing the challenge of training distributed non-IID local datasets [51]. The framework's innovation lay in its adaptive Rényi Differential Privacy-based security budget allocation and asynchronous weight-based grouped update algorithm. This approach was validated experimentally, showcasing its capacity for ensuring robust utility in heterogeneous IoT environments. Turning to mobile and IoT devices, a Federated Learning-based continuous authentication mechanism was developed [52]. It emphasized user privacy while enhancing authentication accuracy, marking a step forward in secure and private user verification.

Healthcare also benefited from Federated Learning innovations, with a new system employing cryptographic methods and a data-quality-based contribution rate to protect patient data [53]. Its application in face recognition within the HAM10000 dataset showed promising outcomes. An innovative Federated Learning system incorporated a reputation mechanism with differential privacy protection [6]. Utilizing both blockchain and mobile edge computing, it aimed to maximize user privacy and accuracy in testing. An incentive mechanism was also designed, rewarding the participation and contribution within the FL ecosystem.

A Federated Learning privacy protection scheme tailored for edge computing (PPFLEC) has also been introduced [54]. This scheme included a lightweight privacy security protocol, digital signature algorithm, and periodic average training to secure gradient privacy without compromising accuracy or succumbing to various cyber threats. Lastly, an algorithm was devised to stratify users based on their privacy requirements [55]. It underwent a comprehensive theoretical analysis and experimental evaluation, featuring an adaptive gradient clip scheme and an enhanced composition method. These innovations aimed to refine noise reduction and clipping, thereby ensuring secure and efficient Federated Learning.

A pioneering framework named VREFL has been developed, advancing the field of Federated Learning by integrating secure aggregation with features that ensure both verifiability and reconnection efficiency [56]. Utilizing homomorphic hash proofs along with CRT for the efficient updating of outdated masks, the VREFL framework has proven secure through rigorous testing. Its effectiveness has been confirmed through extensive trials with real-world datasets and in practical settings.

In the realm of fault recognition within Permanent Magnet Synchronous Motors (PMSMs), a new research initiative has introduced a stacking model that leverages Federated Learning together with enhanced validation strategies [57]. The model's refinement

process involves an advanced version of Particle Swarm Optimization, which fine-tunes the weight coefficients of clients for optimal performance. The resulting dynamic authentication model not only excels in handling data imbalances but also operates at a reduced communication cost, as demonstrated through meticulous experimentation.

2.4. Resource Management

Resource management in Federated Learning (FL) is critical to ensure efficient utilization of computational, communication, and energy resources across distributed devices. Since FL often involves resource-constrained devices like smartphones or IoT devices, effective strategies are needed to manage their participation without overloading them. Techniques like participant selection optimize which devices are involved in each training round based on their availability, computational capacity, and network conditions.

With the proliferation of IoT devices, the integration of Software-Defined Networks, mobile device clouds, and micro-edge data centers has become imperative. These innovations laid the groundwork for the proficient caching of user data [58]. In a detailed analysis, the application of Federated Learning for the training of machine learning models was explored, specifically within the purview of edge computing for caching in the IoT landscape. The study not only reviewed contemporary models but also introduced a control algorithm tailored to address many different challenges.

In another insightful study, an online heuristic approach was proposed to meticulously select the most fitting clients for the Federated Learning training process [59]. It was illuminating to discover that this heuristic method augmented accuracy by an impressive 27% when benchmarked against an online random algorithm.

Addressing the challenges of resource-constrained IoT networks, a comprehensive analytical model was developed [60]. This model was insightful as it factored in unreliable parameter transmissions, resource shortages, and node fairness. By introducing the SRA and RSRA schemes, the aim was to bolster the stability of model training. To add, an adaptive local training model was conceptualized to operate within prescribed resource limits. This was substantiated by empirical tests that ascertained the efficacy of the aforementioned scheme.

ELASTIC has emerged as a groundbreaking, energy-efficient distributed machine learning framework tailored for the IoT domain [61]. This innovative approach harnessed the power of adaptable resource management and discerning client selection algorithms. By opting for a selective client approach—fewer in initial iterations and more in the latter—ELASTIC aimed to reduce energy usage among FL clients. This strategy's effectiveness was demonstrated through simulation results.

Venturing into the domain of the Industrial Internet of Things (IIoT), a unique Federated Learning-driven deep reinforcement learning (DRL) algorithm was formulated [62]. This algorithm was geared towards optimizing the task offloading ratio, bandwidth allocation, and power transmission. Enhancing the allocation efficacy of resources in mobile edge computing-assisted IIoT networks showcased a notable reduction in both the normalized system and communication costs. Moreover, in a bid to mitigate latency challenges during training, RaFed was introduced as a potent resource allocation tool for Federated Learning [63]. Deploying the RGB-D dataset in IIoT systems as a benchmark, it was empirically validated that RaFed could successfully reduce latency by a commendable 29.9%.

Addressing the issue of heterogeneity in IoT environments, a study introduced Hierarchical FL (HFL), a technique utilizing semi-synchronous communications alongside a weighted aggregation strategy [64]. Its application to a milling machining scenario provided evidence that it outperformed both fully synchronous and asynchronous FL schemes in effectiveness.

Researchers have taken strides in computational efficiency with the creation of a gradient sparsification framework known as Gradient General Sparsification [65]. This innovation has demonstrated the remarkable ability to preserve top-1 accuracy while sparsifying 99.9% of gradients, presenting a significant advancement in adaptive optimizers.

A novel framework was formulated to enhance the efficiency of Federated Learning in IIoT networks with limited resources [66]. This approach sought to reduce training delays by integrating device scheduling and resource allocation (DDSRA) into a single optimization problem, tackled using the Lyapunov technique. The framework proved superior to standard baselines, excelling in both convergence time and test accuracy.

In the realm of privacy protection, a noise perturbation scheme has emerged, promising to safeguard client data while sustaining both the accuracy and velocity of Federated Learning with stochastic gradient descent [67]. Notably, it achieves this without imposing additional communication or computational overhead.

An inventive secure sharing method for MIoT data has been crafted, utilizing an edge computing framework enriched with Federated Learning and blockchain technology, underpinned by a novel concept of marine environment quality [68]. This method has been shown to bolster data accuracy and reliability while defending data privacy and security. The development of a new scheduling policy within the Communication-Efficient FL (CEFL) framework has marked a leap forward for Federated Learning over wireless IoT networks [69]. The CEFL framework significantly enhanced learning performance and communication efficiency, manifesting in reduced training loss and elevated testing accuracy.

FED GS, a hierarchical cloud–edge–end FL framework, has been tailored for 5G-enabled factories, aiming to perfect industrial FL training [70]. Combining a binary permutation algorithm with a synchronized compound-step training protocol ensures both heightened accuracy and data security.

Propelled by Federated Learning principles, a new value-added service (VAIS) has revolutionized data and energy transfer for machine learning forecasting in cellular networks [71]. With the power to reduce backhaul data needs by up to 70 times and to minimize energy usage compared with traditional cloud-based methods, VAIS stands poised to alleviate network operator costs while enhancing IoT application performance. An FL technique named GWEP, designed for model compression, has addressed the challenges posed by resource-strapped IoT devices in MEC networks [72]. Its superiority in terms of speed, model compression, energy efficiency, and learning performance over baseline algorithms is noteworthy.

Introducing Clustering-based hierarchical and Two-step-optimized FL (CTFed), this strategy has effectively reduced the communication load by clustering clients and sending only representative model updates to the server [73]. Its success lies in its ability to improve training speeds, refine prediction performance, and stabilize networks.

Focusing on telemonitoring systems in IoT, an optimized user assignment and resource allocation solution has emerged [74], leading to improved classification accuracy and reduced communication rounds, thereby amplifying privacy and speed. In addition, a new integration technique, HCFL [75], has surfaced for AI in IoT networks, which significantly reduces communication costs while ensuring rapid convergence across extensive device arrays. Its efficacy has been confirmed through simulations and analytical evaluations.

Moreover, the concept of a digital twin edge network merges digital twin technology with edge computing to construct accurate IoT device models [76], resulting in heightened communication efficiency and reduced energy costs. FedMCCS stands as a multi-criteria-based approach for client selection in Federated Learning [77]. By maximizing client participation and streamlining training and update transmission, it achieves notable reductions in communication rounds and discarded data, optimizing network traffic.

An adaptation of the FedAvg algorithm incorporating distributed Adam optimization and compression tactics has shown promise in edge computing environments [78], achieving desired accuracies in reduced rounds and with fewer data, as demonstrated in experiments with the MNIST and CIFAR-10 datasets.

Lastly, a cross-chain solution addresses the bandwidth and privacy concerns in e-learning by optimizing model training, privacy, model trading, and payments through

a machine learning-based auction dynamic pricing system [79]. The decentralized FL system's efficiency, scalability, and flexibility have been validated within AIoT networks.

2.5. Network Optimization

Network optimization in Federated Learning (FL) focuses on reducing communication overhead and improving data transfer efficiency between participating devices and the central server. Since FL involves frequent exchanges of model updates, minimizing the communication cost is crucial, especially in bandwidth-constrained environments.

In the realm of the Industrial Internet of Things (IIoT), a new service provision framework called BrainIoT has been created [80]. This scheme excels by integrating production data to optimize the network, harnessing both external and internal relations to enhance service prediction precision. Beyond prediction, it adeptly reserves resources for anticipated services, delivering a substantial uplift in operational performance. Venturing into resource allocation within wireless edge networks for Federated Learning (FL), a novel game-theoretic approach dubbed 'FedBarg' has been developed [81]. It caters to the unique challenge of partial IIoT device participation in FL tasks. Simulations validate that FedBarg's strategic resource distribution significantly outperforms the effectiveness of standard benchmarks.

Exploring collaborative efforts in FL, a three-layered architecture tailored for distributed IIoT networks has been introduced [82]. Central to this framework is a resource optimization using a deep multi-agent reinforcement learning (RoF) mechanism, which meticulously selects devices and allocates resources. A device refinement subroutine is incorporated to accelerate convergence. Comparative simulations affirm this architecture's superiority over conventional benchmarks.

The introduction of an adaptive federated optimization algorithm, known as Adp-FedProx, represents a leap forward in distributed machine learning over IoT networks [83]. This algorithm shines in its economical resource usage, slashing energy consumption by a notable 10%. Empirical trials have underscored Adp-FedProx's advantages, particularly in achieving higher training accuracy and quicker convergence when juxtaposed with its predecessor, FedProx.

2.6. Energy-Efficiency

Energy efficiency in Federated Learning (FL) is a key consideration, especially when working with resource-constrained devices like smartphones, IoT sensors, or edge nodes. Federated Learning involves frequent computation and communication, which can quickly deplete battery life or strain limited energy resources. To address this, energy-efficient FL strategies focus on reducing resource consumption without compromising model performance. Techniques like participant selection prioritize devices with sufficient energy reserves for training rounds.

Exploring the frontier of decentralized Federated Learning within the emerging 6G networks, researchers have innovated a data redistribution phase that hinges on a bargaining game for optimization. They designed both centralized and decentralized algorithms to elevate efficiency to its peak [84]. A subsequent comparative analysis, involving simulations and deep neural network (DNN) training on significant datasets, validated the approach.

In another study, a framework dubbed FogFL was conceptualized for IoT projects where time is of the essence and resources are limited [85]. This framework integrates fog nodes as local aggregators within the Federated Learning paradigm, facilitating site-specific information sharing. This integration not only ramps up efficiency but also diminishes the dependency on central servers. Experimental outcomes indicate a remarkable reduction in communication latency and energy demands for edge devices—plummeting by 92% and 85%, respectively. Diving deeper into optimization, a novel strategy aimed at fine-tuning energy consumption and latency in Federated Learning was set forth for fog-enabled IoT networks [86]. The technique capitalizes on the control of CPU frequency and transmission

power. The challenge was framed as a nonlinear programming problem, with an Alternate Direction algorithm crafted to hunt for optimal solutions.

Another enhancement in Federated Learning algorithms comes with the integration of a weight-based proximal term within the local loss function [87]. Specifically tailored for wireless IoT networks, this algorithm targets the minimization of either total energy consumption or completion time. Experimental data have illustrated that this refined algorithm strides ahead of the conventional FedAvg algorithm in terms of performance.

Lastly, an analysis was conducted to carve out a pathway for minimizing energy consumption in relay IoT networks leveraging Federated Learning [88]. This encompassed a comprehensive consideration of device allocation with relays, power transmission, and computation frequency. The simulations backed up a near-optimal solution that demonstrated a frugal energy consumption profile, albeit with a modest uptick in Federated Learning time.

2.7. Reinforcement Learning

Reinforcement learning (RL) and Federated Learning (FL) can intersect to create powerful decentralized learning systems, where RL agents benefit from federated approaches to tackle privacy, scalability, and distributed data challenges. Additionally, RL can enhance FL by optimizing processes such as participant selection, resource allocation, and communication scheduling, adapting dynamically to changing conditions.

A novel Digital Twin-Enabled Industrial (DTEI) architecture has emerged, leveraging Federated Learning to facilitate real-time assessment of industrial components [89]. This system employs deep reinforcement learning to select IIoT components strategically, aiming to minimize data transmission issues while enhancing personal data protection. The introduction of an asynchronous Federated Learning protocol counters the unpredictable behavior of diverse IIoT components, with empirical evidence pointing to a swifter convergence and heightened training accuracy when benchmarked against conventional methodologies.

Moreover, FedAdapt has been introduced, as an adaptive offloading mechanism rooted in reinforcement learning, engineered to expedite the Federated Learning process for IoT devices with constrained computational capabilities [90]. Through rigorous testing on practical testbeds, FedAdapt demonstrated a remarkable reduction in training time by over 50% for average devices, while reducing the training duration for stragglers and the total training time by up to 57%. Impressively, when pitted against traditional FL approaches, it also managed to cut down training time by 40% without compromising accuracy.

In addition, a cutting-edge approach harnessing deep reinforcement learning has been proposed to orchestrate devices and allocate resources in Federated Learning-powered IIoT systems [91]. The objective is to simultaneously reduce latency, diminish energy use, and elevate model precision, thereby streamlining the entire system. It should be noted that Federated Learning often results in suboptimal performance due to its equal treatment of diverse client data. To address this, a new central server utilizing reinforcement learning [92] has been developed to recognize and adapt to client differences, improving outcomes for most users. Tests confirm that this approach outperforms traditional Federated Learning models.

In the field of secure anomaly detection for IIoT, a robust deep reinforcement learning-based model stands out, excelling in quick and accurate anomaly detection [93]. The proposed model's main feature is its dual emphasis on gauging privacy leakage and analyzing action relationships, proving its efficacy and privacy preservation capabilities across a variety of IIoT environments.

Delving into an Industrial IoT framework that assimilates digital twins, researchers have crafted a system that marries nested Federated Learning with a trust-based mechanism [94]. It is further refined by a Lyapunov dynamic deficiency queue and complemented with deep reinforcement learning to foster an asynchronous structure conducive to clustering. Comparative studies validate this system's enhanced accuracy and energy efficiency against existing benchmarks.

2.8. Semi-Supervised

Semi-supervised learning (SSL) and Federated Learning (FL) complement each other by addressing the challenges of data scarcity and privacy in decentralized environments. SSL leverages both labeled and unlabeled data to improve model performance, which is particularly valuable in FL scenarios where labeled data are limited across devices. In federated semi-supervised learning, devices can collaboratively train models using their locally labeled data and the wealth of unlabeled data available, without sharing raw data.

In the realm of smart city applications, the scarcity of labeled data presents a significant challenge. Recognizing this, researchers developed semi-supervised methodologies for wireless edge networks that effectively harness unlabeled data, tailored to the unique attributes of wireless edge networking [95].

In addition, a novel framework known as FedMalDE was crafted, incorporating a knowledge transfer mechanism to deduce the labels for unlabeled samples [96]. The framework utilizes a specially crafted Subgraph Aggregated Capsule Network (SACN) for the accurate detection of various forms of malicious behavior. When tested with real-world datasets, the framework proved to be effective in identifying malware and maintaining robustness. Furthering the advancements in semi-supervised learning, another approach that combined knowledge distillation with a convolutional neural network was introduced to lower the communication overhead for IoT edge device intrusion detection [97]. Upon evaluation with a real-world traffic dataset, this method demonstrated superior performance over previously established methods.

3. Federated Learning: A Game-Changer Across Multiple Domains in the IoT

Federated Learning (FL) is reshaping numerous domains within the Internet of Things (IoT), demonstrating its versatility from agriculture and healthcare to smart cities, manufacturing, and transportation. In agriculture, FL enhances productivity and security through smart analytics, while in healthcare, it safeguards patient privacy and elevates diagnostic accuracy. Smart homes and transportation systems benefit from FL through improved energy management and security protocols. Manufacturing and energy sectors see advancements in operational efficiency and sustainability. Retail applications utilize FL for insightful consumer analysis and supply chain improvements. In smart cities, FL supports intelligent infrastructure and services, and in education and communications, it fosters personalized experiences and efficient data handling. The adoption of FL in drones further extends its reach into environmental monitoring and advanced security, highlighting FL's significant role in driving IoT innovation while prioritizing privacy and efficiency.

Table 2 below highlights some of the key applications of Federated Learning and IoT in a vast array of domains of knowledge.

Table 2. Applications of Federated Learning and IoT in Various Domains.

Application	Reference(s)
Agriculture	[23,31,98–105]
Healthcare and Hospitals	[5,53,106–115]
Smart Homes	[116–124]
Transportation	[125–137]
Manufacturing	[13,20,21,36,62,63,70,80,93,94,138–142]
Energy	[88,143–151]
Retail/Market	[152–158]
Smart Cities	[159–168]

Table 2. *Cont.*

Application	Reference(s)
Education	[169]
Communications	[27,35,78,79,170–175]
Drones	[12,176–184]

3.1. Agriculture

Federated Learning (FL) has transformative potential in the agriculture domain, enabling collaborative data analysis without compromising data privacy. One key application is precision agriculture, where FL models help optimize crop yield by analyzing distributed data from farms, such as soil conditions, weather patterns, and crop health, without transferring sensitive farm data to a central server. FL can also enhance pest and disease detection by aggregating insights from various regions, enabling early detection and mitigation strategies. In livestock management, FL facilitates the development of models for animal health monitoring, tracking growth patterns, and predicting outbreaks of diseases by utilizing data from multiple farms.

FELIDS emerges as a Federated Learning-based intrusion detection system tailored for agricultural IoT infrastructures, enhancing data privacy through local device learning [23]. Devices contribute to collective intelligence by sharing model updates with a central server for aggregation, thus bolstering the overall detection capabilities without compromising privacy. Tested across several datasets, including CSE-CIC-IDS2018, MQTTset, and InSDN, FELIDS outshines traditional methods, offering an impressive blend of privacy and precision in attack detection.

Moreover, the introduction of PEFL, a deep privacy-encoding Federated Learning framework, marks a significant step toward addressing security and privacy concerns in smart agriculture [98]. By employing a perturbation-based encoding in tandem with an LSTM-autoencoder, followed by an encoded-data-driven federated Gated Recurrent Unit algorithm, PEFL distinguishes itself in accurately identifying attack patterns, showcasing its superiority over conventional models on the ToN-IoT dataset.

The Edge-IoAT concept connects a myriad of smart devices for distributed computing, where a novel Federated Learning framework efficiently processes both vertically and horizontally partitioned data [99]. Complemented by an energy-aware device scheduling algorithm, the framework ensures optimal resource allocation, resulting in fast convergence, low communication costs, and high modeling precision, as evidenced by field tests in Nebraska's agricultural landscape.

In addition, an innovative Amendable Multi-Function Sensor Control (AMFSC) system is on the horizon, designed to enhance agricultural production by smartly adjusting sensor operations [100]. Leveraging Federated Learning, the system continuously updates and validates sensor controls against optimal historical data, ensuring maximum crop development and adaptability. Performance evaluations reveal significant gains in productivity metrics and reductions in operational delays.

PPFchain stands out as a novel blockchain framework infused with Federated Learning to secure sensor-based IoT architectures [101]. By harnessing Federated Learning and cryptographic measures within a lightweight, performance-optimized architecture, the system delivers enhanced security, accuracy, and efficiency. Comparative analysis with traditional blockchain setups demonstrates PPFchain's elevated performance in a distributed environment.

It should also be noted that QuAsyncFL offers a novel solution by blending asynchronous Federated Learning with an innovative quantizer, tackling communication inefficiencies in AIoT systems [102]. Comprehensive analyses and experimental results underline significant strides in communication efficiency, with the model showcasing robust convergence within a specified bound. Confronting Federated Learning's inherent challenges, a

new framework leveraging blockchain technology emerges as a bulwark against potential threats such as single-point failures and various forms of attacks [103]. Empirical evaluations of the Ethereum network underscore the enhanced security, illustrating the combined strength of blockchain and Federated Learning mechanisms.

An innovative approach in IoT intrusion detection emphasizes the privacy of local model training and inference, allowing devices to collaborate in enhancing the detection model with shared updates [31]. Extensive evaluations using the NSL-KDD dataset confirm the approach's ability to safeguard privacy while delivering accuracy that rivals centralized methods, representing a significant advancement in collaborative security techniques.

A cutting-edge platform harnesses the diversity of sensors embedded in smartphones to serve a multitude of smart city applications [104]. This distributed sensing system efficiently prompts devices to contribute data as needed, minimizing the necessity for new installations. By enabling remote data collection and promoting localized decision-making, the system offers a cost-effective and intelligent solution for smart city infrastructures.

In smart agriculture scenarios demanding instantaneous data analysis, a novel strategy for UAV-aided MEC systems efficiently manages NFV-enabled nodes [105]. This strategy addresses the critical balance between energy consumption and timely data processing, employing Federated Learning to navigate these challenges. Simulations demonstrate the strategy's efficacy in maintaining real-time application support while optimizing energy use across the network.

3.2. Healthcare and Hospitals

Federated Learning (FL) has transformative applications in healthcare, enabling privacy-preserving collaboration across institutions without sharing sensitive patient data. In medical imaging analysis, FL allows hospitals to collaboratively train models for detecting diseases like cancer or retinal disorders using X-rays, MRIs, and other imaging data.

In a significant stride towards enhancing healthcare data privacy, a study introduced a Deep Federated Learning (DFL) framework tailored for IoT-based systems [107]. Notably, the framework was employed in an experiment to detect skin conditions. The outcomes were promising, indicating the DFL's adeptness at safeguarding data privacy, elevating model accuracy to an impressive 97%, and minimizing operational costs for service providers.

Exploring the integration of Federated Learning with blockchain, another research presented a secure architecture specifically designed for smart healthcare solutions [106]. Such a combination promises the dual benefits of ensuring data privacy and security. What is more, this approach negates the need for users to share personal data while training machine learning models, paving the way for Federated Learning applications in secure infrastructure across smart cities.

On the technical forefront, a novel quantization algorithm, named FedMSQE, has been introduced for Internet of Medical Things (IoMT) devices [185]. This algorithm focuses on minimizing quantization error in learning frameworks. Comparative studies, including single-node and Federated Learning evaluations, indicated its superiority over existing algorithms, both in terms of accuracy and error reduction.

In the same vein, researchers have proposed the Federated Learning Driven IoMT (FLDIoMT) framework [186]. This comprehensive structure not only promises flexibility in IoMT services but also underscores the importance of privacy and security. Complementing the framework, the authors showcased an efficient data processing workflow, highlighting its practicality with an innovative sleep monitoring device, named 'iSmile'.

Lastly, delving into disease diagnosis, a recent endeavor leveraged Federated Learning in tandem with IoMT and a Variational Autoencoder (VAE) [187]. The goal was to augment patient privacy during the diagnosis process. Specifically targeting coronary heart disease, the mechanism's efficacy was tested using the MIT-BIH arrhythmia database, revealing promising results for patient privacy and diagnosis precision.

In the realm of smart healthcare, secure architecture emerges as a pivotal concern, especially when incorporating Blockchain and IoT cloud platforms [106]. These technologies are crucial in safeguarding security and privacy. Scalable machine learning applications, such as those found in healthcare, are increasingly leveraging Federated Learning. This approach enables users to access well-trained machine learning models without transmitting personal data to the cloud. Discussing the utility of Federated Learning, the technology shows promise for creating a distributed and secure environment within smart cities.

Deep Federated Learning (DFL) frameworks are carving out a niche in healthcare data monitoring and analysis, especially when integrated with IoT devices [107]. An innovative FL algorithm fine-tunes the process of local training data acquisition. A notable experiment in this domain is the use of such a framework to detect skin diseases, yielding impressive outcomes. DFL models have been shown to preserve data privacy effectively, uphold the decentralized system inherent to IoT devices, enhance model performance as evidenced by the area under the curve (AUC) reaching 97%, and concurrently decrease the operational costs for service providers.

When applied to deep learning medical models within IoT-based healthcare systems, Federated Learning introduces an additional layer of privacy [53]. By employing cryptographic tools like masks and homomorphic encryption, the framework fortifies local models against attacks aimed at extracting private medical data. The novel approach of valuing dataset quality over size for determining the contribution rate to the global model each training epoch marks a significant departure from traditional deep learning practices. A resilient dropout-tolerant scheme has been conceptualized, ensuring the Federated Learning process remains unaffected as long as the number of online clients doesn't fall below a certain threshold. Security analysis confirms the scheme's adequacy in preserving data privacy, with theoretical discussions on computational and communication costs. Skin lesion classification experiments, utilizing the HAM10000 dataset, have showcased the scheme's potential to deliver promising results while maintaining privacy.

In addition, an advanced Federated Learning framework has been designed to train deep neural networks, partitioning the network between IoT devices and a centralized server [108]. Here, most of the training computation is managed by the server. By sparsifying activations and gradients, the framework substantially slashes communication overhead. Empirical studies back the efficiency of this framework, suggesting it maintains high accuracy with a minimal synchronization traffic requirement compared with traditional Federated Learning approaches. Exploring Alzheimer's disease detection, a new system called ADDetector harnesses IoT devices to collect audio within smart home environments, implementing novel topic-based linguistic features to refine accuracy [109].

ADDetector employs a unique three-layer architecture for privacy, alongside Federated Learning and differential privacy mechanisms to safeguard raw data integrity and the confidentiality of classification models. The system's novel asynchronous privacy-preserving aggregation framework strengthens the model aggregation process between clients and the cloud. Performance evaluations, including 1010 AD detection trials, underscore ADDetector's proficiency, boasting an accuracy of 81.9% and minimal time overhead, even with all privacy-preserving mechanisms in play.

The limitations of wearable IoT devices present significant challenges in data privacy, service integrity, and network structure adaptability [5]. A new platform using Federated Learning and private blockchain technology within a fog-IoT network addresses these concerns. This innovative structure adapts to wearable IoT devices' needs, with a testbed demonstrating the platform's effectiveness in maintaining patient privacy and predictive service integrity. Further exploration of additional technologies contributes to the security and adaptability of the IoT network, illustrating the platform's capacity to tackle key issues in predictive healthcare.

For decentralized healthcare systems, a deep Federated Learning framework ensures the privacy of users in a distributed setup [110]. The framework includes an algorithm for automated training data acquisition, enhancing its practicality for skin disease detection

and incorporating transfer learning to overcome the limited availability of healthcare data for deep learning models. Results reveal an increase in model performance metrics, including accuracy, precision, recall, and F1-score, while also highlighting the trade-off in terms of model conversion time for users. Despite this, the Federated Learning system fulfills the imperative of building models without compromising private data.

The heterogeneity in participation willingness among users in an FL network presents unique challenges, especially in healthcare applications where sustained user engagement is critical [111]. A dynamic contract design introduces a two-period incentive mechanism that maintains intertemporal incentive compatibility. This innovative approach outperforms uniform pricing schemes and effectively counters the adverse impacts of information asymmetry.

An alternative to Federated Average, FedImpPSO, enhances the robustness of Federated Learning algorithms by utilizing an advanced Particle Swarm Optimization method [112]. This innovation improves the data exchange process and, when evaluated using medical datasets, showcases significant accuracy improvements. The approach proves particularly effective in the context of healthcare, with case studies demonstrating notable accuracy in COVID-19 classification and cardiovascular disease prediction.

In pursuit of energy-efficient solutions for healthcare applications, the Energy-Efficient Distributed Federated Learning Offloading and Scheduling (EDFOS) system in blockchain-enabled IoT networks introduces a model for computational offloading and task scheduling [113]. This model optimizes power consumption and computational performance, with simulation results highlighting its superiority over other current approaches in reducing energy usage and operational time. The EDFOS system's implications for energy and privacy-conscious healthcare applications are substantial, with ongoing research to address any residual challenges.

Edge cloudlets have created new opportunities for Federated Learning applications, particularly in Alzheimer's detection. The hierarchical federated structure proposed in [114] (FEDTIMEDIS) allows for collaboration across disparate computing hierarchies. By segmenting datasets based on disease pathology and leveraging a novel LSTM approach, the model's training is enhanced. Practicality is evidenced through a remote monitoring case study, showcasing the integration of Federated Learning models in edge cloudlets for Alzheimer's detection without necessitating direct data sharing.

The FedHome framework signifies a leap forward in in-home health monitoring, offering a compromise between user privacy and the benefits of cloud-based global model learning [115]. A Generative Convolutional Autoencoder (GCAE) forms the backbone of this framework, which facilitates a balanced dataset for health insights tailored to individual needs. GCAE's lightweight design is instrumental in cutting down communication costs during the Federated Learning process. Empirical experiments confirm the FedHome framework's superior performance over traditional methodologies, marking it as a highly promising avenue for future exploration.

3.3. Smart Homes

Federated Learning (FL) plays a pivotal role in enhancing smart home technologies by enabling personalized and secure machine learning solutions. In voice assistants and IoT devices, FL allows for personalized interactions, such as learning user preferences for automation routines or improving speech recognition, all while keeping user data on the local device. Additionally, FL aids in predictive maintenance by aggregating data from various smart devices to forecast potential failures and schedule timely repairs.

In the field of smart home applications, a novel method called FedAvg [116] enhances Federated Averaging by incorporating a unique weighting scheme that prioritizes local models from smart home devices with recent data. This approach dynamically refines the global prediction model based on data timeliness. When evaluated against other machine learning models using real-world smart home energy consumption data, the method demonstrated improved accuracy, reflected by a lower mean absolute error compared with

standard Federated Averaging, underscoring its efficiency in handling time-sensitive data without compromising privacy.

Building on advancements in computation and communication within edge networks, a newly designed architecture leverages federated edge learning to maintain data privacy in smart homes [117]. This architecture employs a selective process for weighing and dropping nodes based on data quality, optimizing the training process and reducing communication costs. Long Short-Term Memory networks are utilized to forecast future tasks and preemptively store them locally, minimizing redundant computations. Simulations using real-world data reveal the architecture's effectiveness in significantly reducing iterations and cutting down computational and communicational overhead while preserving user privacy.

Extending the application of Federated Learning in smart homes, a multi-level Federated Learning approach utilizes dual house data for comprehensive user behavior analysis through an array of sensors [118]. To address bandwidth issues caused by data heterogeneity, a Federated Learning compression method is applied, resulting in a system that achieves lower latency and more effective interactions. This approach not only enhances accuracy but also outperforms traditional reinforcement learning methods, highlighting its robustness and reliability in handling complex smart home scenarios.

Further enhancing privacy and reliability, an innovative blockchain-based Federated Learning method has been introduced, leveraging edge nodes to form a decentralized blockchain and minimizing risks associated with single points of failure [119]. By training models with local data from IoT devices in homes, the method ensures efficient learning while preserving data privacy. A novel clustering strategy has been developed to address challenges associated with Non-Independent and Identically Distributed data, significantly boosting model accuracy and expanding its applicability across various use cases.

In parallel, a framework tailored for automatic learning of contextual access control policies in smart homes addresses challenges such as data scarcity and the diversity of IoT device usage patterns [120]. This framework, known as LoFTI, employs federated multi-task learning to extract context-aware policies while safeguarding user privacy. By identifying six universal feature types to represent contextual access patterns and implementing a custom data augmentation strategy, LoFTI effectively addresses data imbalance, achieving notable reductions in both false negatives and positives and enhancing the reliability of access control mechanisms.

To further strengthen privacy in communication, a protocol called PPTrans has been developed, forming the foundation for a Federated Learning-based framework, FLYFDetect [121]. This framework supports PPTrans and integrates a Federated Learning strategy built around the YOLOv5 algorithm. FLYFDetect's effectiveness has been demonstrated in real-world environments, particularly in applications like flame detection systems, showcasing its viability for smart home use while ensuring robust privacy protections.

G-Fedfilt emerges as a graph signal processing-inspired aggregation rule that allows structured information flow based on a graph's topology, effectively capturing the interconnectedness of CIoT devices for domain-specific model training [122]. The inclusion of a tunable graph filter parameter offers a continuum between domain-agnostic and domain-specific Federated Learning, with an intrinsic clustering capability based on connectivity. Furthermore, the scheme introduces a communication-efficient scheduling mechanism that significantly reduces computation load and latency, as evidenced by simulations showing enhanced classification accuracy and a remarkable decrease in communication and computation demands.

A federated IoT interactive threat detection system, FedINT, has been introduced to analyze potential interactive threats in smart homes [123]. By representing device interactions as graphs and utilizing graph neural networks encoded within a layer-wise clustering-based federated framework, FedINT permits collaborative GNN model training without exposing raw data. It addresses data heterogeneity and concept drift issues that are

particular to graph data, with datasets from various IoT platforms validating that FedINT surpasses existing methods with an impressive accuracy rate in detecting interactive threats.

Lastly, a new paradigm for deep model training in smart homes is explored, where edge computing takes center stage, enabling local training of deep models and keeping sensitive data within a trusted environment [124]. This Federated Learning collaboration between edge and cloud servers ensures privacy-preserving training of deep models, bringing the benefits of AI technology to smart homes. The scheme's feasibility has been affirmed through experimental validation, promising a privacy-centric approach to AI in smart environments.

3.4. Transportation

Federated Learning (FL) offers innovative applications in the transportation sector, enhancing efficiency, safety, and sustainability. One significant use case is in autonomous vehicles, where FL enables collaborative learning across vehicles to improve navigation, object detection, and driving behavior models while maintaining the privacy of sensitive location and sensor data.

An exploration into an intelligent transportation system named the Road Safety Intelligent Transport System (RSITS) reveals its integration of deep Federated Learning within fog cloud networks to improve road safety alerts [125]. RSITS employs mobile and vehicle-mounted LiDAR sensors, training on vast pedestrian and vehicle detection datasets across multiple fog servers, with central cloud aggregation. The Deep Federated Learning Scheduling Scheme (DFLSS) prioritizes emergency tasks locally for prompt responses while deferring non-critical tasks to the cloud, enforcing strict security protocols at each node through data encryption and decryption.

Building on the technological advancements in intelligent transportation systems, the significance of Software-Defined Networking (SDN) technology is highlighted in its support for the Internet of Vehicles (IoV), enhancing traffic management and road safety [126]. An SDN-enabled IoV architecture is discussed, incorporating trust assessments for nodes to establish a robust framework for an Intrusion Detection System (IDS). Simulations affirm the efficacy of this framework in securing the IoV network and ensuring reliable communication among connected vehicles.

Extending the scope of Federated Learning applications, an analysis of driver behavior demonstrates the collaborative training of Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) and Convolutional Neural Network-Bidirectional LSTM (CNN-Bi-LSTM) models by vehicles [127]. These models classify driver behaviors using real-time data from smartphones and onboard devices, trained without exchanging raw data. Effectiveness is validated on both Independent and Identically Distributed (IID) and non-IID datasets. Further enhancing smart city infrastructure, the development of a blockchain and Federated Learning-enabled framework is outlined for traffic management [128]. This framework offers real-time monitoring capabilities and addresses security concerns in a decentralized manner, improving the efficiency of local and global models while demonstrating resilience to cyberattacks.

A Hierarchical Federated Learning (HFL) based model tailored for anomaly detection within Vehicle IoT (V-IoT) networks combines decision tree techniques with HFL [129], improving the accuracy of malicious activity detection, supported by practical V-IoT scenarios. For traffic flow prediction, the Federated Learning-based Gated Recurrent Unit neural network algorithm (FedGRU) contrasts with centralized methods [130], updating universal models through secure parameter aggregation to protect privacy. It employs a Federated Averaging algorithm to reduce communication overhead and a novel clustering approach for scalability, achieving near-state-of-the-art prediction accuracy.

Expanding on these diverse applications, a Federated Learning framework for geographical Point of Interest (POI) recommendation prioritizes user privacy in distributed networks [131]. Tackling the optimization problem through advanced mathematical techniques ensures only gradient information is shared with the federated server, thus pro-

tecting user data, as shown by real-world dataset experiments. Further showcasing the flexibility of Federated Learning, the Federated Deep Learning based on Spatial–Temporal Long and Short-Term Networks (FedSTNs) algorithm [132] is applied to traffic flow prediction. Utilizing an Attentive Mechanism to federate short-term information under Vertical Federated Learning, its efficacy is confirmed through practical datasets.

Extending the application of Federated Learning to regional transportation, a Federated Learning model for traffic flow prediction is introduced, focusing on controlling vehicle routing and departure times via intelligent autonomous vehicles to optimize travel efficiency [133]. A novel road weight measurement method enhances the framework, with real-time data contributing to reduced average travel times, thereby demonstrating the dual benefits of improved privacy and traffic efficiency.

Moreover, a novel Federated Learning architecture is presented to reduce the transmission load in data exchanges while enhancing security and privacy, supported by a hybrid blockchain architecture [134]. An asynchronous learning approach leverages deep reinforcement learning for efficient node selection, showing improvements in learning accuracy and model convergence.

In vehicle cooperative positioning (CP), a Federated Learning system (FedVCP) is introduced [135], harnessing the Social Internet of Things and collaborative edge computing for privacy-preserving high-precision positioning corrections. Vehicle-to-vehicle communication and transfer learning adjust for individual vehicle differences, with extensive real-data testing confirming its effectiveness over baseline methods.

Moreover, a two-tier Federated Learning model tailored for 6G environments has been proposed [136], integrating a novel selection and aggregation scheme that uses contextual network information to optimize resource allocation and energy consumption. The compilation of studies offers a panoramic view of Federated Learning's impact on creating safer, more efficient, and user-oriented transportation networks.

Lastly, a Federated Learning-based license plate recognition framework (FedLPR) is developed to address privacy and data centralization concerns [137]. This framework is structured to operate on mobile devices, where it utilizes data locally available on these devices to train detection and recognition models through Federated Learning techniques. The design choice ensures that user data remain on the individual's device, rather than being transmitted to a central server. Comprehensive testing of FedLPR illustrates that it achieves high accuracy levels in recognition tasks and maintains a reasonable communication overhead, all while safeguarding user privacy.

3.5. Manufacturing

A key application is in predictive maintenance, where FL aggregates data from machinery in multiple factories to build robust models for detecting equipment failures and scheduling timely interventions without sharing proprietary sensor data. In quality control, FL enables the development of models that analyze production data across facilities to identify defects and optimize manufacturing processes, improving efficiency and reducing waste.

To mitigate issues surrounding anomaly detection in the Industrial Internet of Things (IIoT), a robust strategy employing Federated Learning has been introduced [93]. This strategy leverages deep reinforcement learning (DRL) to cultivate a universal anomaly detection model across devices without needing to share local datasets, thus enhancing privacy.

Digital twins have been integrated into the Industrial IoT architecture, improving Federated Learning processes by mirroring the characteristics of industrial devices [94]. A trust-based aggregation approach compensates for the potential deviations in digital twin estimations, while adjustments to the aggregation frequency, guided by Lyapunov dynamic deficit queue and deep reinforcement learning, aim to refine learning under tight resource availability.

Furthermore, a Federated Learning (FL) framework enables decentralized devices to collaboratively enhance a Deep Anomaly Detection (DAD) model's generalization

capabilities [138]. This is augmented by a Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) model adept at detecting anomalies, supplemented by a gradient compression method to curtail communication costs.

Moreover, an innovative approach for managing mobile edge computing (MEC) in IIoT networks is presented, utilizing a Federated Learning algorithm powered by deep reinforcement learning [62]. This algorithm optimizes task offloading, bandwidth allocation, and transmit power to minimize system costs and communication overheads during optimization. In addition, a resource allocation strategy, named RaFed, has been devised to reduce training latency within Federated Learning frameworks, addressing the challenge through heuristic device selection to balance interference with convergence times [63]. An IIoT system employing an RGB-D dataset demonstrates that RaFed cuts latency considerably.

A multi-tentacle Federated Learning framework introduces a novel way to maintain data trust within Software-Defined IIoT by grouping similar learning tasks and implementing algorithms [20] to detect and minimize poisoning attacks while preserving privacy through differential privacy measures. A chameleon hash scheme with a changeable trapdoor offers a novel security dimension for Federated Learning in IIoT [13]. This method not only undergoes rigorous security analysis but also integrates into a redactable medical blockchain, showing promise in accuracy and efficiency.

By capitalizing on digital twins for an enhanced IIoT environment, a collaborative learning architecture adjusts the frequency of data aggregation based on device performance and resource constraints [139], leading to superior learning accuracy and energy conservation compared with traditional benchmarks. A dispersed Federated Learning framework (DFL) focuses on optimizing resource distribution, solving complex problems through sub-problem simplification and an iterative approach [140]. This strategy outperforms random resource allocation and association in terms of cost-efficiency.

In the same vein, an ensemble-based deep Federated Learning model targets cyberthreat hunting without necessitating data sharing [21]. This dual-component model employs classifier ensembles to reach decisions, outshining its counterparts in F1-score metrics and demonstrating stability and speed against varying client numbers. Digital twin systems have been leveraged within IIoT to fine-tune the relationship between physical and virtual realms, optimizing structure [141]. An advanced communication-efficient algorithm enhances Federated Learning by revamping training updates, which has proven its merit through intelligent camera detection simulations.

Moreover, the use of Autoencoder-based Federated Learning has been proposed in [142] along with vibration data for decentralized training on edge devices, bolstering privacy and network resilience. Trials with real-world datasets confirm the method's robust performance and notable reductions in resource and network demands. The hierarchical framework FED GS for 5G industries improves Federated Learning performance on Non-Independent and Identically Distributed data by employing a gradient-based algorithm and a compound-step synchronization protocol [70]. This approach offers marked improvements in accuracy and efficiency over traditional models.

BrainIoT introduces a novel service provisioning scheme for IIoT, merging industrial knowledge graphs, Federated Learning predictions, and globally optimized resource reservations [80]. This scheme excels in making precise service predictions and optimizing the quality of service. Lastly, a quantization-based Federated Learning mechanism within a device-to-device network emphasizes model convergence [36]. An adaptive stopping mechanism and a synchronization protocol ensure smooth transitions, proving that even with bandwidth-saving quantizers, convergence remains unaffected.

3.6. Energy

FL allows energy providers to collaboratively develop models that predict energy demand, optimize grid performance, and integrate renewable energy sources without sharing customer-specific usage data. In the energy sector, a resource allocation framework aimed at reducing cumulative energy consumption within relay-assisted Internet of

Things (IoT) networks engaged in Federated Learning (FL) has been proposed [88]. The framework focuses on minimizing energy usage by IoT devices while adhering to the FL time constraint, which encompasses both model training computation and wireless transmission delays. Addressing the complexity of this NP-hard optimization challenge, the study formulates joint near-optimal and simpler suboptimal solutions based on graph theory. Comparative simulations demonstrate that the near-optimal method significantly reduces energy consumption compared with traditional fixed computation and power adaptation approaches, albeit with a slight increase in FL time.

Building on efforts to enhance energy efficiency, a joint optimization approach has been proposed that splits the optimization problem into two segments [143]. One segment focuses on IoT device scheduling and power allocation, while the other addresses computation frequency allocation. Conflict graph-based methodologies are iteratively employed to resolve these subproblems, yielding substantial energy savings. This approach proves particularly effective in managing larger networks and data volumes, favoring local model training on IoT devices over Fog Access Points (F-APs), thereby optimizing energy consumption in resource-intensive scenarios.

Expanding on energy-efficient clustering techniques, an energy-and-distribution-aware cooperative clustering algorithm (EDA-CCA) has been introduced to optimize the selection of Cluster Heads (CHs) among IoT devices [144]. Devices with ample energy are designated as CHs, with other devices clustered based on proximity to these CHs and the base station, as well as data distribution. This hierarchical parameter aggregation approach achieves high model accuracy with minimal energy consumption, outperforming alternative schemes and confirming its effectiveness in energy-constrained IoT networks.

In the context of energy security, the FedDetect framework has been developed as a privacy-preserving Federated Learning solution for detecting energy theft [145]. Detection stations employ a local differential privacy scheme for secure data processing, complemented by a homomorphic encryption-based secure protocol for training parameter exchange. Extensive experiments with real energy consumption data validate the utility of the framework's deep learning model, a temporal convolutional network (TCN), which achieves high detection accuracy while reducing computational overhead, thereby ensuring both security and efficiency.

A further advancement is presented in an energy-efficient Federated Learning framework tailored for Digital Twin (DT)-Enabled Industrial Internet of Things (IIoT) [146]. This framework empowers IIoT devices to select optimal training methods based on environmental conditions, facilitating energy-efficient FL either locally or via DT on server resources. A deep reinforcement learning (DRL)-based algorithm addresses the joint problem of training method selection and resource allocation. Simulations underscore substantial energy savings achieved by the framework while maintaining the required convergence rate for FL, showcasing its potential in industrial applications.

Furthermore, a distributed framework leveraging the computational, communicational, and intelligent capacities of edge devices is showcased, aiming to facilitate secure energy trading and network monitoring [147]. By integrating blockchain with Federated Learning alongside advanced network technologies, the framework enables efficient energy transactions and remote monitoring. Its application is particularly impactful within vehicular networks, where reliability and responsiveness are paramount, ensuring robust and adaptive system performance.

Building on privacy and communication optimization, an energy predictor tailored for net-metering systems is introduced [148]. This framework incorporates a hybrid deep learning model for accurate power consumption and generation forecasting. To secure FL training, a novel encryption method, an efficient inner-product functional encryption (IPFE) is employed. Additionally, a change-and-transmit (CAT) strategy enhances communication efficiency by reducing data transmission. Predictive accuracy and improved communication efficiency are validated through extensive studies, underscoring the system's practicality in real-world energy prediction scenarios.

Extending Federated Learning frameworks for heterogeneous networks, an energy-efficient hierarchical Federated Learning (HFL) system supported by wireless energy transfer (WET) is proposed [149]. The framework addresses the resource management problem as a mixed-integer nonlinear programming (MINLP) challenge, introducing a heuristic algorithm, H2RMA, to balance energy consumption, channel quality, and accuracy requirements. The studies highlight the framework's adaptability, confirming its ability to optimize network energy consumption while accounting for device mobility and heterogeneous conditions, thereby enhancing overall network efficiency.

Moreover, a novel client selection mechanism, dubbed "EnerGy-AwaRe CliEnt SElection for Green Federated Learning (GREED)," has been proposed [150]. The algorithm optimizes the balance between maximizing client participation and minimizing energy consumption from client batteries. Through extensive simulations, GREED's performance is validated, demonstrating its efficacy in optimizing client selection while maintaining energy constraints and model upload deadlines.

Lastly, a game theory-based incentive mechanism is proposed for ensuring collaborative security within an energy blockchain environment, particularly for FL [151]. The mechanism encourages a security-centric approach among nodes during iterative FL training. The study establishes that the mechanism can guide the system to a Nash equilibrium, fostering collaborative security. Further evaluations affirm the mechanism's robustness, reliability, and simplicity within the energy blockchain context.

3.7. Retail/Market

Federated Learning (FL) is transforming the retail and market sectors by enabling collaborative insights while preserving consumer privacy. One major application is in personalized recommendations, where FL allows retailers to improve recommendation systems by learning from user behavior across stores or platforms without accessing individual customer data directly. In inventory management, FL aggregates sales data from distributed locations to enhance demand forecasting, reducing overstocking and shortages.

A novel collaborative clustering technique is introduced that eschews the need for exchanging raw data by implementing an attention-based model within a Federated Learning setup [152]. Edge devices execute model computations using their local data and relay the updates to a centralized aggregator until a point of convergence is reached. The model utilizes transaction data to train and produce a concise embedding. Once convergence is achieved, the model is disseminated among clients or stores, and a clustering-based dynamic method (CBDM) is then applied to segment customers according to their purchasing behaviors, with semantic embedding employed for pattern extraction. This approach has demonstrated commendable performance, evidenced by 0.75 and 0.70 ROC values for random and fixed distributions, respectively, all while promoting privacy and reducing communication overhead.

Moreover, a forecasting framework for retail supply chain commodity demand harnesses vertical Federated Learning to overcome prevailing data security and privacy challenges [153]. The approach integrates datasets from diverse platforms—including social media, e-commerce, and retail—within a shared region, with results underscoring the enhanced performance compared with existing algorithms. An innovative bidding application for the wholesale power market, named FedLPQS, merges Federated Learning with blockchain technology [154]. This application brings together cost data from several power plants for increased transparency and records these details on a blockchain platform, specifically tailored for consortium chains. The expected outcome is to bring down costs in wholesale power transactions.

A distributed method for pinpointing consumer characteristics utilizes Federated Learning to safeguard retailer privacy [155]. The method applies privacy-conscious principal component analysis (PCA) to distill smart meter data features, which are then used to train an artificial neural network under a Federated Learning regime. Case studies

conducted with the Irish CER dataset affirm that the method yields results on par with centralized data processing techniques.

An FL-based deep learning model is crafted to identify characteristics of residential households, a crucial aspect for retailers aiming to tailor services to customers [156]. The model combines convolutional neural networks with long short-term memory networks to capture spatial-temporal features from energy consumption patterns, operating under a decentralized FL framework. An advanced asynchronous stochastic gradient descent algorithm with delay compensation is introduced, aiming to boost training speed and model accuracy.

In response to privacy concerns around smart meter data belonging to different entities, two Federated Learning approaches are unveiled for the extraction of electricity consumption patterns [157]. These approaches facilitate the k-means clustering algorithm's distributed training, preserving data ownership integrity. The approaches' validity is backed by experiments on real-world smart meter datasets, illustrating Federated Learning's aptitude for handling sensitive data.

The FedREP framework emerges as a new method for energy load forecasting for Retail Electricity Providers (REPs), focusing on preserving privacy within a horizontal Federated Learning context [158]. This system uses differential privacy to allow multiple REPs to collectively create a robust machine learning model without data exchange. An LSTM neural network is employed for its advanced time series forecasting capabilities, with rigorous testing on actual energy consumption data demonstrating that the framework achieves privacy and scalability and maintains performance levels in close range to those of centralized models.

3.8. Smart Cities

FL facilitates real-time optimization of traffic lights and congestion prediction by aggregating data from sensors and vehicles without centralizing sensitive location information. It supports public safety through improved models for crime prediction and emergency response by leveraging distributed data from surveillance cameras and IoT devices.

Addressing the concerns of data fragmentation and high trust costs in smart city IoT systems, a lightweight and trustworthy sharing mechanism is introduced [159]. This mechanism harnesses the synergy between blockchain and Federated Learning for secure data sharing and instills trust in the process. Further refining the system, a credit-value-based node selection algorithm and a smart contract-based node evaluation algorithm are devised to enhance the learning quality. An optimized raft consensus is also developed to meet the smart city consortium blockchain's unique demands for delay sensitivity and security. Simulation assessments of the respective algorithms reveal that the mechanism holds significant promise for application, showing an uptick in the accuracy of the Federated Learning model, albeit with increased training times.

To counteract the issues arising from data diversity in smart city environments, a multi-tier Federated Learning mechanism is structured [160]. Reinforcement learning agents are employed to select nodes, counterbalancing the effects of non-IID data distributions. Moreover, an asynchronous, non-blocking model aggregation and update process is utilized to free up resources on faster devices, hence optimizing the efficiency and stability of the learning process. Simulation outcomes point to the efficacy of this method, enhancing the performance of Federated Learning tasks within the densely device-populated edge networks of smart cities.

The robustness of the FE-BDP algorithm is confirmed as it consistently maintains accuracy levels above 95%, showing minimal fluctuation despite the increasing number of users [161]. This stability is coupled with the FE-LDP model exhibiting a notably lower loss value than comparative models and maintaining this advantage over time. Blockchain encryption significantly reduces data aggregation times to under a second. Furthermore, an edge-asynchronous communication framework proves to be versatile across various

urban applications, facilitating effective data communication with substantial accuracy improvements and reduced communication overhead.

Leveraging data from ubiquitous devices in the urban landscape, AI models are trained to bolster smart services benefiting society broadly [162]. Applying Lyapunov optimization theory, a dynamic queue model is formulated for each edge node's online dataset. An innovative online task assignment algorithm, known as the Federated Online Control Algorithm (FedOCA), is brought forth. Empirical studies indicate a notable accuracy improvement over traditional Fedavg algorithms on the FashionMNIST dataset, with further analysis shedding light on batch size impacts on performance.

A novel FL-based BGRU model has been engineered to accurately forecast air quality using large-scale data from smart cities [163]. Its performance is evaluated against other machine learning models through various metrics, establishing it as a valuable tool in environmental monitoring and management. Expanding on health applications, a novel architecture for medical IoT is proposed, designed to automate the sensing of health and environmental parameters for assessing obesity risk [164]. This architecture integrates BMI readings with an advanced Federated Learning system, with MATLAB simulations analyzing the proposed algorithm. Observations on loss function trends suggest the potential of FL to recommend personalized fitness routines, paving the way for improved patient care and reduced chronic disease risks.

Advancing the efficiency of Federated Learning models, a Particle Swarm Optimization technique is introduced to refine the hyperparameter settings of local machine learning models within a collaborative learning architecture [165]. Its efficacy is demonstrated through two case studies: smart city traffic prediction and Industrial IoT predictive maintenance. This technique achieves a significant reduction in communication rounds required for hyperparameter tuning, maintaining high precision when optimizing configurations for both federated and centralized learning models, thereby enhancing scalability and efficiency in diverse applications.

Additionally, a federated personalization approach is explored, emphasizing collaborative knowledge exchange among individual learners to enrich local specificities without compromising shared generalization [166]. This framework is exemplified through real-world deployment at a French computing heater company, where edge devices enable secure and sustainable ambient intelligence. The approach proves instrumental in reducing generalization loss across tasks, such as time series forecasting and image classification, through the effective transfer of inter-learner knowledge, demonstrating its potential in achieving balance between local adaptation and global learning.

Further enhancing Federated Learning capabilities, an innovative data-sharing method merges blockchain technology with Federated Learning to improve the reliability of smart city operations [167]. Blockchain ensures the integrity of node performance data, enabling the development of a node selection algorithm and an IPoQ consensus incentive mechanism to streamline Federated Learning tasks. Additionally, the integration of differential privacy technology strengthens resistance to differential attacks, confirming the effectiveness of this approach in establishing a secure and efficient data-sharing model for Federated Learning in complex smart city environments.

To boost the efficacy and reliability of traditional federated recommendation systems, a new semi-asynchronous hierarchical system, HFSA, is suggested [168]. This system introduces an edge server layer, mitigating communication loads on central servers and enabling more frequent global aggregation. It encompasses a semi-asynchronous aggregation mechanism that adapts to the pace of client updates, accounting for device heterogeneity and ensuring model convergence. The system's performance, as demonstrated through comparative baseline analyses, reveals its potential in delivering enhanced recommendation services with superior accuracy and reduced training durations, alongside an evaluation of key influential factors.

3.9. Education

Federated Learning (FL) is revolutionizing education by enabling data-driven insights while protecting student and institutional privacy. The exploration of smart classroom pedagogy begins with an examination of the foundational principles and unique features of IoT technology [169]. The effectiveness of these educational approaches is evaluated through practical implementation in the context of smart classroom settings for teaching Chinese. Furthermore, to gain insight into the impact of these strategies, a random survey involving questionnaires was administered to 150 students majoring in Chinese in a certain city. Findings from the survey suggest that smart classrooms serve as a dynamic communication hub, enabling both teachers and students to organize and revisit educational content effectively. However, challenges arise in engaging students' enthusiasm fully due to individual differences in perspective and initiative, leading to a deviation in the anticipated outcomes of smart language classroom teaching. The study provides a theoretical and practical framework for the development of innovative smart teaching models, marking a significant stride in the advancement of smart educational methodologies.

3.10. Communications

In the field of communications, a novel General Gradient Sparsification (GGS) framework is introduced to refine the update process of sparse gradients for adaptive optimizers [170]. This framework incorporates two critical mechanisms: gradient adjustments to enhance convergence and the update of batch normalization (BN) layers using local gradients to mitigate delays in gradient updates without increasing communication demands. Testing across network architectures such as LeNet-5 and AlexNet demonstrates that even with a substantial 99.9%.

Expanding on advancements in Federated Learning, a Distributed Federated Learning (DFL) framework facilitated by an Intelligent Reflecting Surface (IRS) enhances aerial-terrestrial networks without relying on a central server [171]. Model parameters are communicated to satellites post-training, with the IRS dynamically adjusting the wireless propagation environment to optimize resource usage. An algorithm designed to maximize the secrecy rate improves system effectiveness by up to 20%, offering a more efficient alternative to the traditional Semi-Definite Programming (SDP) algorithm. Similarly, a pioneering Federated Learning approach leverages 6G technology through a super-wireless-over-the-air methodology [35]. By utilizing interference-resistant communication channels, this system ensures secure and high-performance AI training, significantly advancing IoT applications in terms of intelligence, breadth, and speed.

Bridging the gap between physical network infrastructure and digital representations, the integration of digital twins with edge networks gives rise to Digital Twin Edge Networks (DITENs) [172]. To bolster communication security and data privacy, a blockchain-powered Federated Learning scheme is adopted. This scheme features an asynchronous aggregation protocol coupled with a reinforcement learning strategy, optimizing user relay scheduling and spectrum allocation. Simulations confirm that this integrated approach significantly enhances communication efficiency and data security, making it a robust solution for IoT applications.

Adding to that, an Efficient Adaptive Federated Optimization (FedEAFO) algorithm is introduced to improve the efficiency of Federated Learning by minimizing learning errors [173]. The algorithm jointly optimizes variables such as local updates and parameter compression, enabling fine-tuning to balance computation, communication, and accuracy. Experimental comparisons demonstrate that FedEAFO achieves higher accuracy levels at a faster pace compared with leading algorithms, validating its effectiveness in resource-constrained settings.

Advancements in Federated Learning techniques also include an adaptation of FedAvg, which incorporates a distributed form of Adam optimization to reduce the number of convergence rounds required [78]. When combined with innovative compression techniques, this communication-efficient FedAvg (CE-FedAvg) achieves significant reductions

in communication rounds and data uploads. Testing on datasets such as MNIST and CIFAR-10 reveals that CE-FedAvg maintains desired accuracy levels even under aggressive compression, demonstrating resilience and enhanced performance, particularly in real-time applications like those on a Raspberry Pi testbed.

In the realm of anomaly detection, a deep anomaly detection framework is tailored for time series data sensing within the Industrial Internet of Things (IIoT) [27]. Utilizing an attention mechanism-based convolutional neural network paired with long short-term memory (AMCNN-LSTM) units, the framework excels in capturing salient features and predicting time series anomalies. A gradient compression mechanism, based on Top-k selection, enhances communication efficiency, halving communication overhead compared with traditional FL frameworks, while ensuring timely and accurate anomaly detection.

Furthermore, a cross-chain framework is developed for efficient and flexible model training in Federated Learning [79]. Each blockchain is designed to handle specific FL tasks with a focus on privacy, and a cross-chain mechanism facilitates seamless collaboration for model training, exchange, and transactions. Complementing this, a model update compression scheme conserves communication resources with minimal accuracy loss, while a dynamic pricing strategy rooted in machine learning algorithms supports auction-based model trading. The results confirm the framework's scalability and efficiency across AIoT platforms.

The introduction of FedQNN signifies a breakthrough in Federated Learning frameworks, particularly for IoT applications requiring efficient computation and communication [175]. This framework integrates ultra-low-bitwidth quantization, enabling lightweight fixed-point computations with reduced energy consumption. Combining sparsification and quantization, FedQNN establishes itself as a powerful tool for energy-efficient Federated Learning, addressing the unique challenges of IoT environments.

3.11. Drones

Investigations into the power management of drones within a Federated Learning (FL) system have led to the development of an algorithm that maximizes the security rate of the system [176]. This is achieved by balancing the battery capacities of the drones and the quality of service requirements, such as FL training time. The algorithm, which addresses this as a non-linear programming problem, has been validated through comprehensive simulations, showcasing its low computational complexity and efficiency.

Building on this, advancements in semi-supervised Federated Learning have been tailored specifically for UAV image recognition tasks [177]. A novel parameter mixing strategy named FedMix has been implemented to overcome the challenges posed by heterogeneous local data in terms of quantity, features, and distribution. Additionally, an aggregation rule, termed FedFreq, has been introduced to mitigate statistical heterogeneity by weighting local models based on the frequency of client participation in the training process.

Further extending the role of UAVs in Federated Learning systems, they are deployed as aerial intermediaries for model aggregation and update relay [174]. This approach enhances FL's reach to data contributors in unreliable network conditions, improving communication efficiency. To facilitate UAV service provider participation, a multi-dimensional contract incentive design ensures that packages align with operational costs, enabling seamless integration and scalability.

Complementing these developments, efficiency in processing road-related data has seen improvements through an FL-based computing paradigm over dew computing, primarily utilized in Internet of Drones (IoD) applications like parking space discovery [178]. Experiments with traffic datasets affirm the viability of this innovative approach, highlighting its potential to significantly enhance application performance.

Expanding on the convergence of emerging technologies, a survey on the integration of FL and blockchain technologies has been conducted, focusing on enabling drone edge intelligence for environmentally sustainable practices [179]. This study explores the technological synergy, motivations, and frameworks that underpin this convergence, paving the

way for smart, green environments while identifying future opportunities and challenges in the domain.

To address privacy and security concerns, a decentralized FL framework incorporates methods for maximizing the secrecy rate through user and radio resource block scheduling [180]. By reframing the scheduling issue as a maximum-weight independent set problem using graph theory, efficient resolutions are achieved. Moreover, distributed schemes have been explored to reduce reliance on centralized policies. Enhancing this trend, a swarm of drones in a wireless network utilizes deep learning models, such as ResNet50 coupled with U-net, to achieve high pixel accuracy in semantic image segmentation tasks [181].

Shifting the focus to communication optimization, a new data-driven stochastic framework has been introduced to address gaps in existing Multi-path TCP models [182]. This framework integrates intelligent learning agents to optimize packet management and beamforming, presenting a cross-layer architecture informed by multi-armed bandits and Federated Learning. Building on this, a Multi-level Asynchronous Federated Learning architecture has been developed to streamline data sharing and training across various entities. This architecture adapts strategies for drones, edge entities, and central entities, enhancing performance, convergence speed, and training accuracy within the IoD infrastructure [183].

Further emphasizing energy efficiency, a novel algorithm has been designed to adjust the CPU frequencies of drones, minimizing energy consumption during FL training while adhering to time constraints [184]. Extensive simulation results validate the efficacy of this approach, marking a significant step in sustainable IoD network management. Lastly, integrating blockchain technology with Federated Transfer Learning, the FTLchain architecture has emerged to elevate cybersecurity in collaborative AI without compromising privacy [12]. By employing a hierarchical consensus mechanism, FTLchain ensures scalability and robustness, confirmed through detailed simulations, showcasing its potential in secure, decentralized AI applications.

In recent studies, UAV swarms integrated with CNN-LSTM and distributed Federated Learning have been applied to predict AQI in the Delhi region [127]. Remarkably, these models showcased superior accuracy compared with several other temporal methodologies. Another intriguing research focused on aquatic environments introduced a MADDPG-based algorithm tailored for Federated Learning computation in conjunction with the Internet of Underwater Things (IoUT) devices [188]. This method surpassed the regular actor–critic and DDPG algorithms, especially concerning downlink throughput.

Addressing the concerns of data privacy in distributed sensing networks, researchers designed a Federated Learning model that updates asynchronously [189]. The outcomes were promising, displaying not only high accuracy but also reduced convergence time and bandwidth overhead relative to prevalent techniques. Tackling network scarcity and cybersecurity challenges in remote regions with scarce IoT devices, a novel Federated Learning-based Data Accumulation Scheme has been proposed [190]. Utilizing drones and blockchain technology, this approach emphasized a dual-phase authentication procedure that bolstered data privacy and protection. Following rigorous testing, the blockchain successfully archived the final model after acquiring approval from the respective mining nodes.

Furthermore, the integration of AI-based models in the Internet of Things landscape has surged in popularity [174]. Recognizing the potential of Federated Learning, a strategy was advanced where model training occurs without the data ever leaving their original user. A UAV-assisted system emerged from this line of thought, promising enhanced communication efficacy. What is more, this model proposes enticing vendors through lucrative incentives, such as covering travel expenditures.

4. Discussion

Communication efficiency is a fundamental challenge in Federated Learning (FL) due to the inherent need for frequent exchange of model updates between distributed clients and the central server. Unlike traditional centralized machine learning, FL requires repeated iterations of transmitting gradients or model weights, which can impose significant communication overhead, particularly in large-scale deployments involving thousands or millions of devices. This is further compounded by the variability in network bandwidth, latency, and reliability among participating clients. For instance, edge devices like smartphones and IoT sensors often operate in environments with intermittent connectivity or limited data transfer capabilities, making it difficult to maintain consistent communication during training.

One major source of inefficiency is the size of model updates, which can be particularly large in deep learning models with millions of parameters. Transmitting these updates in their raw form during every training round can quickly overwhelm the network, leading to bottlenecks, increased latency, and even dropped updates. This is especially problematic in cross-device FL scenarios where the participating devices may have constrained resources and operate over slow or costly wireless networks.

To address these issues, FL incorporates several strategies for optimizing communication efficiency. Model compression techniques, such as quantization, sparsification, and pruning, reduce the size of updates by approximating or selectively transmitting model parameters without significantly compromising accuracy. For example, sparsification involves sending only the most significant gradients, while discarding less critical ones. Similarly, update frequency reduction, where clients locally train the model for multiple epochs before sharing updates, minimizes the number of communication rounds. This approach, often referred to as local updating, balances communication overhead and model convergence speed. Another key strategy is client selection, where only a subset of available clients participate in each training round based on factors like network conditions, device availability, or data representativeness. By reducing the number of active participants, FL systems can optimize communication without sacrificing model quality. Additionally, asynchronous communication protocols allow clients to transmit updates at their own pace, accommodating heterogeneous device capabilities and mitigating delays caused by slower clients.

Data heterogeneity, often referred to as Non-Independent and Identically Distributed (Non-IID) data, is one of the most significant challenges in Federated Learning (FL). Unlike traditional machine learning, where data are typically pooled into a centralized repository and assumed to follow a uniform distribution, FL operates on decentralized datasets that can vary widely across clients. This leads to statistical heterogeneity, where the data distributions, feature spaces, and label distributions differ significantly among clients.

When the local data of clients poorly represent the global data distribution, the updates generated during local training may conflict, resulting in a global model that fails to generalize effectively. This issue is particularly pronounced in scenarios where data imbalance exists, such as when certain classes are overrepresented on some clients and underrepresented or absent on others. Addressing data heterogeneity in FL requires tailored strategies. Personalized Federated Learning (PFL) has emerged as a prominent approach, enabling clients to adapt the global model to their local data distributions. Techniques such as meta-learning or local fine-tuning allow clients to personalize models while benefiting from global collaboration. Additionally, clustered FL partitions clients into groups with similar data distributions, enabling more targeted aggregation and reducing the impact of extreme heterogeneity. Aggregation techniques such as Federated Averaging (FedAvg) have also been adapted to account for data heterogeneity, incorporating weighting schemes that balance the influence of clients based on their data quality or volume. Another promising direction involves transfer learning, where pretrained global models are fine-tuned locally to enhance performance on specific client data.

Scalability is a critical challenge in Federated Learning (FL), especially as it is designed to operate across potentially millions of devices or large-scale organizational networks. While the distributed nature of FL facilitates decentralized training, it also introduces significant computational and communication bottlenecks that impede its scalability. In cross-device FL, where individual clients like smartphones or IoT devices participate, the vast number of devices creates logistical challenges, including selecting which devices will participate in each round, managing their intermittent availability, and ensuring consistent global model updates. These constraints are further exacerbated by the heterogeneous capabilities of devices, ranging from high-performance systems to resource-constrained sensors with limited processing power, memory, and energy.

As the scale of FL increases, the central server can become a bottleneck due to the high volume of updates it must process and aggregate. This problem is particularly acute in scenarios with frequent training iterations or large client populations, as the server's computational and storage requirements grow linearly with the number of participants. To address these scalability and computational issues, several strategies have been proposed. Client selection algorithms optimize the subset of devices participating in each training round based on factors like data representativeness, device availability, and computational capacity. Hierarchical FL architectures introduce intermediate aggregators, such as regional servers, to distribute the computational load and reduce communication traffic to the central server. These intermediate layers perform localized model aggregation before transmitting updates to the central server, significantly improving scalability in large networks.

Table 3 presents some widely used FL frameworks, along with their features.

Table 3. Federated Learning Frameworks.

Framework	Developer	Key Features	Use Cases	Link
TensorFlow Federated (TFF)	Google	Open-source, built on TensorFlow, simulates FL scenarios, tools for custom FL algorithms.	Research and development in TensorFlow.	www.tensorflow.org/federated (accessed on 6 November 2024)
PySyft	OpenMined	Privacy-preserving ML, remote tensor operations, supports PyTorch and TensorFlow.	Privacy-preserving FL, research, prototyping.	github.com/OpenMined/PySyft (accessed on 1 October 2024)
Flower	Adap	Framework-agnostic, easy APIs for FL, supports TensorFlow, PyTorch, JAX.	Multi-framework FL experiments, scalable research.	flower.ai/
FATE (Federated AI Technology Enabler)	WeBank AI	Industrial-grade FL, secure computing, horizontal and vertical FL, big data integration.	Healthcare, finance, IoT applications.	github.com/FederatedAI/FATE (accessed on 6 November 2024)
NVIDIA Clara FL	NVIDIA	Focused on healthcare, tools for edge device orchestration, supports secure aggregation.	Medical and healthcare FL applications.	developer.nvidia.com/industries/healthcare (accessed on 6 November 2024)
FedML	FedML AI Lab	Large-scale FL, supports cloud-edge-device computing architecture, simulation tools.	IoT, edge computing, and cross-device scenarios.	fedml.ai/home

Table 3. *Cont.*

Framework	Developer	Key Features	Use Cases	Link
OpenFL	Intel	Modular FL framework, tools for FL pipelines, supports secure aggregation.	Secure and scalable FL in enterprise environments.	openfl.readthedocs.io/en/latest/ (accessed on 6 November 2024)
PaddleFL	Baidu	Built on PaddlePaddle, supports horizontal/vertical/transfer FL, includes encryption modules.	Industrial applications (especially in China).	github.com/PaddlePaddle/PaddleFL (accessed on 6 November 2024)

5. Future Perspectives

As technology continues to advance and machine learning models grow in complexity, the prevalence of Federated Learning (FL) and Internet of Things (IoT) technologies is poised to rise significantly. The expansive network of IoT devices provides a fertile ground for FL, enabling real-time data collection and processing across myriad nodes. This symbiotic relationship between FL and IoT is expected to catalyze a new wave of technological advancements, driving smarter, more autonomous systems that are both scalable and adaptable to the evolving landscape of digital innovation.

Many IoT devices are constrained by their reliance on battery power and limited resources, making the computational demands of Federated Learning lead to increased energy use. Developing Federated Learning algorithms that are energy-efficient and suited for IoT devices is thus crucial. In environments like vehicular or mobile sensor networks where device participation can change rapidly, with devices frequently entering or exiting the network, there's a significant challenge in adjusting Federated Learning algorithms to handle such dynamic changes smoothly.

Energy consumption and environmental concerns also arise due to the distributed nature of FL, which requires significant computational resources and communication between devices. Mitigating these impacts requires efficient protocols and consideration of the carbon footprint associated with FL deployments. Lastly, transparency in how FL models are trained, evaluated, and deployed is essential to maintaining trust and ensuring alignment with societal values. Addressing these challenges responsibly will require a multi-disciplinary effort encompassing technical innovation, robust governance frameworks, and continuous dialogue among stakeholders.

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References

1. Krishnan, S.; Anand, A.J.; Srinivasan, R.; Kavitha, R.; Suresh, S. *Federated Learning*; CRC Press: Boca Raton, FL, USA, 2024.
2. Yadav, S.P.; Bhati, B.S.; Mahato, D.P.; Kumar, S. *Federated Learning for IoT Applications*; Springer: Berlin/Heidelberg, Germany, 2022.

3. Zhang, C.; Xu, Y.; Elahi, H.; Zhang, D.; Tan, Y.; Chen, J.; Zhang, Y. A blockchain-based model migration approach for secure and sustainable Federated Learning in iot systems. *IEEE Internet Things J.* **2022**, *10*, 6574–6585. [CrossRef]
4. Xu, Y.; Lu, Z.; Gai, K.; Duan, Q.; Lin, J.; Wu, J.; Choo, K.K.R. Besifl: Blockchain empowered secure and incentive Federated Learning paradigm in iot. *IEEE Internet Things J.* **2021**, *10*, 6561–6573. [CrossRef]
5. Baucas, M.J.; Spachos, P.; Plataniotis, K.N. Federated Learning and blockchain-enabled fog-IoT platform for wearables in predictive healthcare. *IEEE Trans. Comput. Soc. Syst.* **2023**, *10*, 1732–1741. [CrossRef]
6. Zhao, Y.; Zhao, J.; Jiang, L.; Tan, R.; Niyato, D.; Li, Z.; Lyu, L.; Liu, Y. Privacy-preserving blockchain-based Federated Learning for IoT devices. *IEEE Internet Things J.* **2020**, *8*, 1817–1829. [CrossRef]
7. Cheng, X.; Tian, W.; Shi, F.; Zhao, M.; Chen, S.; Wang, H. A blockchain-empowered cluster-based Federated Learning model for blade icing estimation on iot-enabled wind turbine. *IEEE Trans. Ind. Inform.* **2022**, *18*, 9184–9195. [CrossRef]
8. Otoum, S.; Al Ridhawi, I.; Mouftah, H. Securing critical IoT infrastructures with blockchain-supported Federated Learning. *IEEE Internet Things J.* **2021**, *9*, 2592–2601. [CrossRef]
9. Zhang, W.; Lu, Q.; Yu, Q.; Li, Z.; Liu, Y.; Lo, S.K.; Chen, S.; Xu, X.; Zhu, L. Blockchain-based Federated Learning for device failure detection in industrial IoT. *IEEE Internet Things J.* **2020**, *8*, 5926–5937. [CrossRef]
10. Kalapaaking, A.P.; Khalil, I.; Rahman, M.S.; Atiquzzaman, M.; Yi, X.; Almashor, M. Blockchain-based Federated Learning with secure aggregation in trusted execution environment for internet-of-things. *IEEE Trans. Ind. Inform.* **2022**, *19*, 1703–1714. [CrossRef]
11. Salim, S.; Turnbull, B.; Moustafa, N. A blockchain-enabled explainable Federated Learning for securing internet-of-things-based social media 3.0 networks. *IEEE Trans. Comput. Soc. Syst.* **2021**, *11*, 4681–4697. [CrossRef]
12. Wang, H.; Jiang, H.; Wu, J.; Zhou, P.; Huang, H. Blockchain-Governed Federated Transfer Learning for Secure Internet of Drones Networks. *IEEE Internet Things Mag.* **2022**, *5*, 134–139. [CrossRef]
13. Wei, J.; Zhu, Q.; Li, Q.; Nie, L.; Shen, Z.; Choo, K.K.R.; Yu, K. A redactable blockchain framework for secure Federated Learning in Industrial Internet of Things. *IEEE Internet Things J.* **2022**, *9*, 17901–17911. [CrossRef]
14. Abubaker, Z.; Javaid, N.; Almogren, A.; Akbar, M.; Zuair, M.; Ben-Othman, J. Blockchained service provisioning and malicious node detection via Federated Learning in scalable Internet of Sensor Things networks. *Comput. Netw.* **2022**, *204*, 108691. [CrossRef]
15. Jin, H.; Dai, X.; Xiao, J.; Li, B.; Li, H.; Zhang, Y. Cross-cluster Federated Learning and blockchain for internet of medical things. *IEEE Internet Things J.* **2021**, *8*, 15776–15784. [CrossRef]
16. Li, S.; Ngai, E.; Voigt, T. Byzantine-robust aggregation in Federated Learning empowered industrial iot. *IEEE Trans. Ind. Inform.* **2021**, *19*, 1165–1175. [CrossRef]
17. Ma, X.; Jiang, Q.; Shojafar, M.; Alazab, M.; Kumar, S.; Kumari, S. DisBezant: secure and robust Federated Learning against byzantine attack in IoT-enabled MTS. *IEEE Trans. Intell. Transp. Syst.* **2022**, *24*, 2492–2502. [CrossRef]
18. Zhang, J.; Ge, C.; Hu, F.; Chen, B. RobustFL: Robust Federated Learning against poisoning attacks in industrial IoT systems. *IEEE Trans. Ind. Inform.* **2021**, *18*, 6388–6397. [CrossRef]
19. Sánchez, P.M.S.; Celdrán, A.H.; Schenk, T.; Iten, A.L.B.; Bovet, G.; Pérez, G.M.; Stiller, B. Studying the robustness of anti-adversarial Federated Learning models detecting cyberattacks in iot spectrum sensors. *IEEE Trans. Dependable Secur. Comput.* **2022**, *21*, 573–584. [CrossRef]
20. Li, G.; Wu, J.; Li, S.; Yang, W.; Li, C. Multitentacle Federated Learning over software-defined industrial internet of things against adaptive poisoning attacks. *IEEE Trans. Ind. Inform.* **2022**, *19*, 1260–1269. [CrossRef]
21. Jahromi, A.N.; Karimipour, H.; Dehghantanha, A. An ensemble deep Federated Learning cyber-threat hunting model for Industrial Internet of Things. *Comput. Commun.* **2023**, *198*, 108–116. [CrossRef]
22. Abdel-Basset, M.; Moustafa, N.; Hawash, H. Privacy-preserved cyberattack detection in industrial edge of things (IEOT): a blockchain-orchestrated Federated Learning approach. *IEEE Trans. Ind. Inform.* **2022**, *18*, 7920–7934. [CrossRef]
23. Friha, O.; Ferrag, M.A.; Shu, L.; Maglaras, L.; Choo, K.K.R.; Nafaa, M. FELIDS: Federated Learning-based intrusion detection system for agricultural Internet of Things. *J. Parallel Distrib. Comput.* **2022**, *165*, 17–31. [CrossRef]
24. Weinger, B.; Kim, J.; Sim, A.; Nakashima, M.; Moustafa, N.; Wu, K.J. Enhancing IoT anomaly detection performance for Federated Learning. *Digit. Commun. Netw.* **2022**, *8*, 314–323. [CrossRef]
25. Cui, L.; Qu, Y.; Xie, G.; Zeng, D.; Li, R.; Shen, S.; Yu, S. Security and privacy-enhanced Federated Learning for anomaly detection in IoT infrastructures. *IEEE Trans. Ind. Inform.* **2021**, *18*, 3492–3500. [CrossRef]
26. Mothukuri, V.; Khare, P.; Parizi, R.M.; Pouriyeh, S.; Dehghantanha, A.; Srivastava, G. Federated-learning-based anomaly detection for IoT security attacks. *IEEE Internet Things J.* **2021**, *9*, 2545–2554. [CrossRef]
27. Liu, Y.; Garg, S.; Nie, J.; Zhang, Y.; Xiong, Z.; Kang, J.; Hossain, M.S. Deep anomaly detection for time-series data in industrial IoT: A communication-efficient on-device Federated Learning approach. *IEEE Internet Things J.* **2020**, *8*, 6348–6358. [CrossRef]
28. Rey, V.; Sánchez, P.M.S.; Celdrán, A.H.; Bovet, G. Federated Learning for malware detection in IoT devices. *Comput. Netw.* **2022**, *204*, 108693. [CrossRef]
29. Ruby, R.; Yang, H.; Wu, K. Anti-jamming strategy for Federated Learning in internet of medical things: A game approach. *IEEE J. Biomed. Health Inform.* **2022**, *27*, 888–899. [CrossRef]
30. Liu, W.; Lin, H.; Wang, X.; Hu, J.; Kaddoum, G.; Piran, M.J.; Alamri, A. D2MIF: A malicious model detection mechanism for Federated Learning empowered artificial intelligence of things. *IEEE Internet Things J.* **2021**, *10*, 2141–2151. [CrossRef]

31. Rahman, S.A.; Tout, H.; Talhi, C.; Mourad, A. Internet of things intrusion detection: Centralized, on-device, or Federated Learning? *IEEE Netw.* **2020**, *34*, 310–317. [CrossRef]
32. Foukalas, F.; Tziouvaras, A. Federated Learning protocols for IoT edge computing. *IEEE Internet Things J.* **2022**, *9*, 13570–13581. [CrossRef]
33. Cui, Y.; Cao, K.; Cao, G.; Qiu, M.; Wei, T. Client scheduling and resource management for efficient training in heterogeneous IoT-edge Federated Learning. *IEEE Trans.-Comput.-Aided Des. Integr. Circuits Syst.* **2021**, *41*, 2407–2420. [CrossRef]
34. Li, J.; Lyu, L.; Liu, X.; Zhang, X.; Lyu, X. FLEAM: A Federated Learning empowered architecture to mitigate DDoS in industrial IoT. *IEEE Trans. Ind. Inform.* **2021**, *18*, 4059–4068. [CrossRef]
35. Pei, J.; Li, S.; Yu, Z.; Ho, L.; Liu, W.; Wang, L. Federated Learning Encounters 6G Wireless Communication in the Scenario of Internet of Things. *IEEE Commun. Stand. Mag.* **2023**, *7*, 94–100. [CrossRef]
36. Ma, T.; Wang, H.; Li, C. Quantized distributed Federated Learning for industrial internet of things. *IEEE Internet Things J.* **2021**, *10*, 3027–3036. [CrossRef]
37. Chen, Y.; Su, Y.; Zhang, M.; Chai, H.; Wei, Y.; Yu, S. FedTor: An Anonymous Framework of Federated Learning in Internet of Things. *IEEE Internet Things J.* **2022**, *9*, 18620–18631. [CrossRef]
38. Savazzi, S.; Nicoli, M.; Rampa, V. Federated Learning with cooperating devices: A consensus approach for massive IoT networks. *IEEE Internet Things J.* **2020**, *7*, 4641–4654. [CrossRef]
39. Mai, T.; Yao, H.; Xu, J.; Zhang, N.; Liu, Q.; Guo, S. Automatic double-auction mechanism for Federated Learning service market in Internet of Things. *IEEE Trans. Netw. Sci. Eng.* **2022**, *9*, 3123–3135. [CrossRef]
40. Chen, Z.; Liao, W.; Hua, K.; Lu, C.; Yu, W. Towards asynchronous Federated Learning for heterogeneous edge-powered internet of things. *Digit. Commun. Netw.* **2021**, *7*, 317–326. [CrossRef]
41. Chen, J.; Xue, J.; Wang, Y.; Huang, L.; Baker, T.; Zhou, Z. Privacy-Preserving and Traceable Federated Learning for data sharing in industrial IoT applications. *Expert Syst. Appl.* **2023**, *213*, 119036. [CrossRef]
42. Wei, Z.; Pei, Q.; Zhang, N.; Liu, X.; Wu, C.; Taherkordi, A. Lightweight Federated Learning for large-scale IoT devices with privacy guarantee. *IEEE Internet Things J.* **2021**, *10*, 3179–3191. [CrossRef]
43. Zeng, Q.; Lv, Z.; Li, C.; Shi, Y.; Lin, Z.; Liu, C.; Song, G. FedProLs: Federated Learning for IoT perception data prediction. *Appl. Intell.* **2023**, *53*, 3563–3575. [CrossRef]
44. Ruzafa-Alcázar, P.; Fernández-Saura, P.; Mármol-Campos, E.; González-Vidal, A.; Hernández-Ramos, J.L.; Bernal-Bernabe, J.; Skarmeta, A.F. Intrusion detection based on privacy-preserving Federated Learning for the industrial IoT. *IEEE Trans. Ind. Inform.* **2021**, *19*, 1145–1154. [CrossRef]
45. Fang, C.; Guo, Y.; Hu, Y.; Ma, B.; Feng, L.; Yin, A. Privacy-preserving and communication-efficient Federated Learning in internet of things. *Comput. Secur.* **2021**, *103*, 102199. [CrossRef]
46. Farahani, B.; Monsefi, A.K. Smart and collaborative industrial IoT: A Federated Learning and data space approach. *Digit. Commun. Netw.* **2023**, *9*, 436–447. [CrossRef]
47. Zhao, Y.; Zhao, J.; Yang, M.; Wang, T.; Wang, N.; Lyu, L.; Niyato, D.; Lam, K.Y. Local differential privacy-based Federated Learning for internet of things. *IEEE Internet Things J.* **2020**, *8*, 8836–8853. [CrossRef]
48. Rahman, A.; Hasan, K.; Kundu, D.; Islam, M.J.; Debnath, T.; Band, S.S.; Kumar, N. On the ICN-IoT with Federated Learning integration of communication: Concepts, security-privacy issues, applications, and future perspectives. *Future Gener. Comput. Syst.* **2023**, *138*, 61–88. [CrossRef]
49. Xu, R.; Chen, Y. μDFL: A Secure Microchained Decentralized Federated Learning Fabric atop IoT Networks. *IEEE Trans. Netw. Serv. Manag.* **2022**, *19*, 2677–2688. [CrossRef]
50. Fu, A.; Zhang, X.; Xiong, N.; Gao, Y.; Wang, H.; Zhang, J. VFL: A verifiable Federated Learning with privacy-preserving for big data in industrial IoT. *IEEE Trans. Ind. Inform.* **2020**, *18*, 3316–3326. [CrossRef]
51. Zhang, T.; Song, A.; Dong, X.; Shen, Y.; Ma, J. Privacy-preserving asynchronous grouped Federated Learning for IoT. *IEEE Internet Things J.* **2021**, *9*, 5511–5523. [CrossRef]
52. Wazzeh, M.; Ould-Slimane, H.; Talhi, C.; Mourad, A.; Guizani, M. Privacy-preserving continuous authentication for mobile and iot systems using warmup-based Federated Learning. *IEEE Netw.* **2022**, *37*, 224–230. [CrossRef]
53. Zhang, L.; Xu, J.; Vijayakumar, P.; Sharma, P.K.; Ghosh, U. Homomorphic encryption-based privacy-preserving Federated Learning in iot-enabled healthcare system. *IEEE Trans. Netw. Sci. Eng.* **2022**, *10*, 2864–2880. [CrossRef]
54. Wang, R.; Lai, J.; Zhang, Z.; Li, X.; Vijayakumar, P.; Karuppiah, M. Privacy-preserving Federated Learning for internet of medical things under edge computing. *IEEE J. Biomed. Health Inform.* **2022**, *27*, 854–865. [CrossRef] [PubMed]
55. Liu, W.; Cheng, J.; Wang, X.; Lu, X.; Yin, J. Hybrid differential privacy based Federated Learning for Internet of Things. *J. Syst. Archit.* **2022**, *124*, 102418. [CrossRef]
56. Ye, H.; Liu, J.; Zhen, H.; Jiang, W.; Wang, B.; Wang, W. VREFL: Verifiable and reconnection-efficient Federated Learning in IoT scenarios. *J. Netw. Comput. Appl.* **2022**, *207*, 103486. [CrossRef]
57. Li, Y.; Chen, Y.; Zhu, K.; Bai, C.; Zhang, J. An effective Federated Learning verification strategy and its applications for fault diagnosis in industrial IOT systems. *IEEE Internet Things J.* **2022**, *9*, 16835–16849. [CrossRef]
58. Balasubramanian, V.; Aloqaily, M.; Reisslein, M.; Scaglione, A. Intelligent resource management at the edge for ubiquitous IoT: An SDN-based Federated Learning approach. *IEEE Netw.* **2021**, *35*, 114–121. [CrossRef]

59. Mohammed, I.; Tabatabai, S.; Al-Fuqaha, A.; El Bouanani, F.; Qadir, J.; Qolomany, B.; Guizani, M. Budgeted online selection of candidate IoT clients to participate in Federated Learning. *IEEE Internet Things J.* **2020**, *8*, 5938–5952. [CrossRef]
60. Li, Z.; Zhou, Y.; Wu, D.; Tang, T.; Wang, R. Fairness-aware Federated Learning with unreliable links in resource-constrained Internet of things. *IEEE Internet Things J.* **2022**, *9*, 17359–17371. [CrossRef]
61. Yu, L.; Albelaihi, R.; Sun, X.; Ansari, N.; Devetsikiotis, M. Jointly optimizing client selection and resource management in wireless Federated Learning for internet of things. *IEEE Internet Things J.* **2021**, *9*, 4385–4395. [CrossRef]
62. Guo, Y.; Zhao, Z.; He, K.; Lai, S.; Xia, J.; Fan, L. Efficient and flexible management for Industrial Internet of Things: A Federated Learning approach. *Comput. Netw.* **2021**, *192*, 108122. [CrossRef]
63. Gao, W.; Zhao, Z.; Min, G.; Ni, Q.; Jiang, Y. Resource allocation for latency-aware Federated Learning in Industrial Internet of Things. *IEEE Trans. Ind. Inform.* **2021**, *17*, 8505–8513. [CrossRef]
64. Herabad, M.G. Communication-efficient semi-synchronous hierarchical federated learning with balanced training in heterogeneous IoT edge environments. *Internet Things* **2023**, *21*, 100642. [CrossRef]
65. Kumar, R.; Khan, A.A.; Kumar, J.; Golilarz, N.A.; Zhang, S.; Ting, Y.; Zheng, C.; Wang, W. Blockchain-federated-learning and deep learning models for covid-19 detection using ct imaging. *IEEE Sens. J.* **2021**, *21*, 16301–16314. [CrossRef] [PubMed]
66. Deng, X.; Li, J.; Ma, C.; Wei, K.; Shi, L.; Ding, M.; Chen, W. Low-Latency Federated Learning With DNN Partition in Distributed Industrial IoT Networks. *IEEE J. Sel. Areas Commun.* **2022**, *41*, 755–775. [CrossRef]
67. Liu, T.; Hu, X.; Xu, H.; Shu, T.; Nguyen, D.N. High-accuracy low-cost privacy-preserving Federated Learning in IoT systems via adaptive perturbation. *J. Inf. Secur. Appl.* **2022**, *70*, 103309. [CrossRef]
68. Qin, Z.; Ye, J.; Meng, J.; Lu, B.; Wang, L. Privacy-preserving blockchain-based Federated Learning for marine Internet of Things. *IEEE Trans. Comput. Soc. Syst.* **2021**, *9*, 159–173. [CrossRef]
69. Chen, H.; Huang, S.; Zhang, D.; Xiao, M.; Skoglund, M.; Poor, H.V. Federated Learning over wireless IoT networks with optimized communication and resources. *IEEE Internet Things J.* **2022**, *9*, 16592–16605. [CrossRef]
70. Li, Z.; He, Y.; Yu, H.; Kang, J.; Li, X.; Xu, Z.; Niyato, D. Data heterogeneity-robust Federated Learning via group client selection in industrial iot. *IEEE Internet Things J.* **2022**, *9*, 17844–17857. [CrossRef]
71. Mian, A.N.; Shah, S.W.H.; Manzoor, S.; Said, A.; Heimerl, K.; Crowcroft, J. A value-added IoT service for cellular networks using federated learning. *Comput. Netw.* **2022**, *213*, 109094. [CrossRef]
72. Prakash, P.; Ding, J.; Chen, R.; Qin, X.; Shu, M.; Cui, Q.; Guo, Y.; Pan, M. Iot device friendly and communication-efficient Federated Learning via joint model pruning and quantization. *IEEE Internet Things J.* **2022**, *9*, 13638–13650. [CrossRef]
73. Zhang, C.; Cui, L.; Yu, S.; James, J. A communication-efficient Federated Learning scheme for IoT-based traffic forecasting. *IEEE Internet Things J.* **2021**, *9*, 11918–11931. [CrossRef]
74. Abdellatif, A.A.; Mhaisen, N.; Mohamed, A.; Erbad, A.; Guizani, M.; Dawy, Z.; Nasreddine, W. Communication-efficient hierarchical Federated Learning for IoT heterogeneous systems with imbalanced data. *Future Gener. Comput. Syst.* **2022**, *128*, 406–419. [CrossRef]
75. Nguyen, M.D.; Lee, S.M.; Pham, Q.V.; Hoang, D.T.; Nguyen, D.N.; Hwang, W.J. Hcfl: A high compression approach for communication-efficient Federated Learning in very large scale iot networks. *IEEE Trans. Mob. Comput.* **2022**, *22*, 6495–6507. [CrossRef]
76. Lu, Y.; Huang, X.; Zhang, K.; Maharjan, S.; Zhang, Y. Communication-efficient Federated Learning for digital twin edge networks in industrial IoT. *IEEE Trans. Ind. Inform.* **2020**, *17*, 5709–5718. [CrossRef]
77. AbdulRahman, S.; Tout, H.; Mourad, A.; Talhi, C. FedMCCS: Multicriteria client selection model for optimal IoT Federated Learning. *IEEE Internet Things J.* **2020**, *8*, 4723–4735. [CrossRef]
78. Mills, J.; Hu, J.; Min, G. Communication-efficient Federated Learning for wireless edge intelligence in IoT. *IEEE Internet Things J.* **2019**, *7*, 5986–5994. [CrossRef]
79. Kang, J.; Li, X.; Nie, J.; Liu, Y.; Xu, M.; Xiong, Z.; Niyato, D.; Yan, Q. Communication-efficient and cross-chain empowered Federated Learning for artificial intelligence of things. *IEEE Trans. Netw. Sci. Eng.* **2022**, *9*, 2966–2977. [CrossRef]
80. Yang, H.; Yuan, J.; Li, C.; Zhao, G.; Sun, Z.; Yao, Q.; Bao, B.; Vasilakos, A.V.; Zhang, J. BrainIoT: Brain-like productive services provisioning with federated learning in industrial IoT. *IEEE Internet Things J.* **2021**, *9*, 2014–2024. [CrossRef]
81. Chakraborty, A.; Misra, S. QoS-aware resource bargaining for Federated Learning over edge networks in industrial IoT. *IEEE Trans. Netw. Sci. Eng.* **2022**, *10*, 2769–2778. [CrossRef]
82. Zhang, W.; Yang, D.; Wu, W.; Peng, H.; Zhang, N.; Zhang, H.; Shen, X. Optimizing Federated Learning in distributed industrial IoT: A multi-agent approach. *IEEE J. Sel. Areas Commun.* **2021**, *39*, 3688–3703. [CrossRef]
83. He, J.; Guo, S.; Qiao, D.; Yi, L. HeteFL: Network-Aware Federated Learning Optimization in Heterogeneous MEC-Enabled Internet of Things. *IEEE Internet Things J.* **2022**, *9*, 14073–14086. [CrossRef]
84. Zhao, J.; Feng, Y.; Chang, X.; Xu, P.; Li, S.; Liu, C.H.; Yu, W.; Tang, J.; Crowcroft, J. Energy-efficient and fair IoT data distribution in decentralised Federated Learning. *IEEE Trans. Netw. Sci. Eng.* **2022**, *10*, 1352–1363. [CrossRef]
85. Saha, R.; Misra, S.; Deb, P.K. FogFL: Fog-assisted Federated Learning for resource-constrained IoT devices. *IEEE Internet Things J.* **2020**, *8*, 8456–8463. [CrossRef]
86. Yao, J.; Ansari, N. Enhancing Federated Learning in fog-aided IoT by CPU frequency and wireless power control. *IEEE Internet Things J.* **2020**, *8*, 3438–3445. [CrossRef]

87. Nguyen, V.D.; Sharma, S.K.; Vu, T.X.; Chatzinotas, S.; Ottersten, B. Efficient Federated Learning algorithm for resource allocation in wireless IoT networks. *IEEE Internet Things J.* **2020**, *8*, 3394–3409. [CrossRef]
88. Al-Abiad, M.S.; Hassan, M.Z.; Hossain, M.J. Energy-Efficient Resource Allocation for Federated Learning in NOMA-Enabled and Relay-Assisted Internet of Things Networks. *IEEE Internet Things J.* **2022**, *9*, 24736–24753. [CrossRef]
89. Yang, W.; Xiang, W.; Yang, Y.; Cheng, P. Optimizing Federated Learning with deep reinforcement learning for digital twin empowered industrial IoT. *IEEE Trans. Ind. Inform.* **2022**, *19*, 1884–1893. [CrossRef]
90. Wu, D.; Ullah, R.; Harvey, P.; Kilpatrick, P.; Spence, I.; Varghese, B. Fedadapt: Adaptive offloading for iot devices in Federated Learning. *IEEE Internet Things J.* **2022**, *9*, 20889–20901. [CrossRef]
91. Zhao, T.; Li, F.; He, L. DRL-based joint resource allocation and device orchestration for hierarchical Federated Learning in NOMA-enabled industrial IoT. *IEEE Trans. Ind. Inform.* **2022**, *19*, 7468–7479. [CrossRef]
92. Pang, J.; Huang, Y.; Xie, Z.; Han, Q.; Cai, Z. Realizing the heterogeneity: A self-organized Federated Learning framework for IoT. *IEEE Internet Things J.* **2020**, *8*, 3088–3098. [CrossRef]
93. Wang, X.; Garg, S.; Lin, H.; Hu, J.; Kaddoum, G.; Piran, M.J.; Hossain, M.S. Toward accurate anomaly detection in Industrial Internet of Things using hierarchical Federated Learning. *IEEE Internet Things J.* **2021**, *9*, 7110–7119. [CrossRef]
94. Sun, W.; Lei, S.; Wang, L.; Liu, Z.; Zhang, Y. Adaptive Federated Learning and digital twin for industrial internet of things. *IEEE Trans. Ind. Inform.* **2020**, *17*, 5605–5614. [CrossRef]
95. Albaseer, A.; Abdallah, M.; Al-Fuqaha, A.; Erbad, A.; Dobre, O.A. Semi-supervised Federated Learning over heterogeneous wireless iot edge networks: Framework and algorithms. *IEEE Internet Things J.* **2022**, *9*, 25626–25642. [CrossRef]
96. Pei, X.; Deng, X.; Tian, S.; Zhang, L.; Xue, K. A knowledge transfer-based semi-supervised Federated Learning for IoT malware detection. *IEEE Trans. Dependable Secur. Comput.* **2022**, *20*, 2127–2143. [CrossRef]
97. Zhao, R.; Wang, Y.; Xue, Z.; Ohtsuki, T.; Adebisi, B.; Gui, G. Semi-supervised Federated Learning based intrusion detection method for internet of things. *IEEE Internet Things J.* **2022**, *10*, 8645–8657. [CrossRef]
98. Kumar, P.; Gupta, G.P.; Tripathi, R. PEFL: Deep privacy-encoding-based Federated Learning framework for smart agriculture. *IEEE Micro* **2021**, *42*, 33–40. [CrossRef]
99. Yu, C.; Shen, S.; Zhang, K.; Zhao, H.; Shi, Y. Energy-aware device scheduling for joint Federated Learning in edge-assisted internet of agriculture things. In Proceedings of the 2022 IEEE Wireless Communications and Networking Conference (WCNC), Austin, TX, USA, 10–13 April 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 1140–1145.
100. Abu-Khadrah, A.; Ali, A.M.; Jarrah, M. An Amendable Multi-Function Control Method using Federated Learning for Smart Sensors in Agricultural Production Improvements. *ACM Trans. Sens. Netw.* **2023**.. [CrossRef]
101. Sezer, B.B.; Turkmen, H.; Nuriyev, U. PPFchain: A novel framework privacy-preserving blockchain-based Federated Learning method for sensor networks. *Internet Things* **2023**, *22*, 100781. [CrossRef]
102. Liu, Y.; Huang, P.; Yang, F.; Huang, K.; Shu, L. QuAsyncFL: Asynchronous Federated Learning with quantization for cloud-edge-terminal collaboration enabled AIoT. *IEEE Internet Things J.* **2023**, *11*, 59–69. [CrossRef]
103. Puri, V.; Solanki, V.K.; Aponte, G.J.R. Blockchain and Federated Learning Based Integrated Approach for Agricultural Internet of Things. In Proceedings of the International Conference on Intelligent Systems & Networks, Hanoi, Vietnam, 18–19 March 2023; Springer: Berlin/Heidelberg, Germany, 2023; pp. 240–246.
104. Imteaj, A.; Amini, M.H. Distributed sensing using smart end-user devices: Pathway to Federated Learning for autonomous IoT. In Proceedings of the 2019 International Conference on Computational Science and Computational Intelligence (CSCI), Las Vegas, NV, USA, 5–7 December 2019; IEEE: Piscataway, NJ, USA, 2019; pp. 1156–1161.
105. Akbari, M.; Syed, A.; Kennedy, W.S.; Erol-Kantarci, M. Constrained Federated Learning for AoI-limited SFC in UAV-aided MEC for Smart Agriculture. *IEEE Trans. Mach. Learn. Commun. Netw.* **2023**, *1*, 277–295. [CrossRef]
106. Singh, S.; Rathore, S.; Alfarraj, O.; Tolba, A.; Yoon, B. A framework for privacy-preservation of IoT healthcare data using Federated Learning and blockchain technology. *Future Gener. Comput. Syst.* **2022**, *129*, 380–388. [CrossRef]
107. Elayan, H.; Aloqaily, M.; Guizani, M. Sustainability of healthcare data analysis IoT-based systems using deep Federated Learning. *IEEE Internet Things J.* **2021**, *9*, 7338–7346. [CrossRef]
108. Yuan, B.; Ge, S.; Xing, W. A Federated Learning framework for healthcare iot devices. *arXiv* **2020**, arXiv:2005.05083.
109. Li, J.; Meng, Y.; Ma, L.; Du, S.; Zhu, H.; Pei, Q.; Shen, X. A Federated Learning based privacy-preserving smart healthcare system. *IEEE Trans. Ind. Inform.* **2021**, *18*, 2021–2031. [CrossRef]
110. Elayan, H.; Aloqaily, M.; Guizani, M. Deep Federated Learning for IoT-based decentralized healthcare systems. In Proceedings of the 2021 International Wireless Communications and Mobile Computing (IWCMC), Harbin, China, 28 June–2 July 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 105–109.
111. Lim, W.Y.B.; Garg, S.; Xiong, Z.; Niyato, D.; Leung, C.; Miao, C.; Guizani, M. Dynamic contract design for Federated Learning in smart healthcare applications. *IEEE Internet Things J.* **2020**, *8*, 16853–16862. [CrossRef]
112. Houssein, E.H.; Sayed, A. Boosted Federated Learning based on improved Particle Swarm Optimization for healthcare IoT devices. *Comput. Biol. Med.* **2023**, *163*, 107195. [CrossRef]
113. Mohammed, M.A.; Lakhan, A.; Abdulkareem, K.H.; Zebari, D.A.; Nedoma, J.; Martinek, R.; Kadry, S.; Garcia-Zapirain, B. Energy-efficient distributed Federated Learning offloading and scheduling healthcare system in blockchain based networks. *Internet Things* **2023**, *22*, 100815. [CrossRef]

114. Gupta, D.; Kayode, O.; Bhatt, S.; Gupta, M.; Tosun, A.S. Hierarchical Federated Learning based anomaly detection using digital twins for smart healthcare. In Proceedings of the 2021 IEEE 7th International Conference on Collaboration and Internet Computing (CIC), Atlanta, GA, USA, 13–15 December 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 16–25.
115. Wu, Q.; Chen, X.; Zhou, Z.; Zhang, J. Fedhome: Cloud-edge based personalized Federated Learning for in-home health monitoring. *IEEE Trans. Mob. Comput.* **2020**, *21*, 2818–2832. [CrossRef]
116. Skianis, K.; Giannopoulos, A.; Gkonis, P.; Trakadas, P. Data aging matters: Federated Learning-based consumption prediction in smart homes via age-based model weighting. *Electronics* **2023**, *12*, 3054. [CrossRef]
117. Nour, B.; Cherkaoui, S.; Mlika, Z. Federated Learning and proactive computation reuse at the edge of smart homes. *IEEE Trans. Netw. Sci. Eng.* **2021**, *9*, 3045–3056. [CrossRef]
118. Ravichandran, P.; Saravanakumar, C.; Rose, J.D.; Vijayakumar, M.; Lakshmi, V.M. Efficient multilevel federated compressed reinforcement learning of smart homes using deep learning methods. In Proceedings of the 2021 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), Chennai, India, 24–25 September 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 1–11.
119. Zhu, B.; Lu, K.; Tao, T. A Blockchain-Based Federated Learning for Smart Homes. In Proceedings of the 2023 4th International Conference on Information Science, Parallel and Distributed Systems (ISPDS), Dalian, China, 12–14 May 2023; IEEE: Piscataway, NJ, USA, 2023, pp. 689–693.
120. Yu, T.; Li, T.; Sun, Y.; Nanda, S.; Smith, V.; Sekar, V.; Seshan, S. Learning context-aware policies from multiple smart homes via federated multi-task learning. In Proceedings of the 2020 IEEE/ACM Fifth International Conference on Internet-of-Things Design and Implementation (IoTDI), Sydney, Australia, 21–24 April 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 104–115.
121. Chen, B.; Liu, Y.; Zhang, S.; Chen, J.; Han, Z. Flyfdetect: A smart home privacy protection framework via federated learning. In Proceedings of the 2022 7th IEEE International Conference on Data Science in Cyberspace (DSC), Guilin, China, 11–13 July 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 432–439.
122. Rasti-Meymandi, A.; Sheikholeslami, S.M.; Abouei, J.; Plataniotis, K.N. Graph Federated Learning for CIoT Devices in Smart Home Applications. *IEEE Internet Things J.* **2022**, *10*, 7062–7079. [CrossRef]
123. Wang, G.; Yan, Q. Federated threat detection for smart home iot rules. In Proceedings of the International Workshop on Federated Learning for Distributed Data Mining, Shanghai, China, 4 December 2023.
124. Dong, Y.; Dai, F.; Qin, M. A Privacy-Preserving Deep Learning Scheme for Edge-enhanced Smart Homes. In Proceedings of the 2022 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCom/CyberSciTech), Falerna, Italy, 12–15 September 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 1–6.
125. Grønli, T.M.; Lakhan, A.; Younas, M. RSITS: Road Safety Intelligent Transport System in Deep Federated Learning Assisted Fog Cloud Networks. In Proceedings of the International Conference on Mobile Web and Intelligent Information Systems, Marrakech, Morocco, 14–16 August 2023; Springer: Berlin/Heidelberg, Germany, 2023; pp. 20–37.
126. Hbaieb, A.; Ayed, S.; Chaari, L. Federated Learning based IDS approach for the IoV. In Proceedings of the 17th International Conference on Availability, Reliability and Security, Vienna, Austria, 23–26 August 2022; pp. 1–6.
127. Chhabra, R.; Singh, S.; Khullar, V. Privacy enabled driver behavior analysis in heterogeneous IoV using Federated Learning. *Eng. Appl. Artif. Intell.* **2023**, *120*, 105881. [CrossRef]
128. Sharma, V.; Seetharaman, T.; Varalakshmi, B.; Khangaonkar, A.M. Blockchain and Federated Learning Enabled Smart Traffic Management System for Smart Cities. In Proceedings of the 2023 4th International Conference on Intelligent Engineering and Management (ICIEM), London, UK, 9–11 May 2023; IEEE: Piscataway, NJ, USA, 2023; pp. 1–6.
129. Gupta, D.; Moni, S.S.; Tosun, A.S. Integration of digital twin and Federated Learning for securing vehicular internet of things. In Proceedings of the 2023 International Conference on Research in Adaptive and Convergent Systems, Gdansk, Poland, 6–10 August 2023; pp. 1–8.
130. Liu, Y.; James, J.; Kang, J.; Niyato, D.; Zhang, S. Privacy-preserving traffic flow prediction: A Federated Learning approach. *IEEE Internet Things J.* **2020**, *7*, 7751–7763. [CrossRef]
131. Huang, J.; Tong, Z.; Feng, Z. Geographical POI recommendation for Internet of Things: A federated learning approach using matrix factorization. *Int. J. Commun. Syst.* **2022**. [CrossRef]
132. Yuan, X.; Chen, J.; Yang, J.; Zhang, N.; Yang, T.; Han, T.; Taherkordi, A. Fedstn: Graph representation driven Federated Learning for edge computing enabled urban traffic flow prediction. *IEEE Trans. Intell. Transp. Syst.* **2022**, *24*, 8738–8748. [CrossRef]
133. Su, Z.; Li, C. On improving the regional transportation efficiency based on Federated Learning. *J. Frankl. Inst.* **2023**, *360*, 4973–5000. [CrossRef]
134. Lu, Y.; Huang, X.; Zhang, K.; Maharjan, S.; Zhang, Y. Blockchain empowered asynchronous Federated Learning for secure data sharing in internet of vehicles. *IEEE Trans. Veh. Technol.* **2020**, *69*, 4298–4311. [CrossRef]
135. Kong, X.; Gao, H.; Shen, G.; Duan, G.; Das, S.K. Fedvcp: A federated-learning-based cooperative positioning scheme for social internet of vehicles. *IEEE Trans. Comput. Soc. Syst.* **2021**, *9*, 197–206. [CrossRef]
136. Zhou, X.; Liang, W.; She, J.; Yan, Z.; Kevin, I.; Wang, K. Two-layer Federated Learning with heterogeneous model aggregation for 6g supported internet of vehicles. *IEEE Trans. Veh. Technol.* **2021**, *70*, 5308–5317. [CrossRef]
137. Kong, X.; Wang, K.; Hou, M.; Hao, X.; Shen, G.; Chen, X.; Xia, F. A Federated Learning-based license plate recognition scheme for 5G-enabled internet of vehicles. *IEEE Trans. Ind. Inform.* **2021**, *17*, 8523–8530. [CrossRef]

138. Liu, Y.; Kumar, N.; Xiong, Z.; Lim, W.Y.B.; Kang, J.; Niyato, D. Communication-efficient Federated Learning for anomaly detection in Industrial Internet of Things. In Proceedings of the GLOBECOM 2020–2020 IEEE Global Communications Conference, Taipei, Taiwan, 7–11 December 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 1–6.
139. Song, Q.; Lei, S.; Sun, W.; Zhang, Y. Adaptive Federated Learning for digital twin driven industrial Internet of Things. In Proceedings of the 2021 IEEE Wireless Communications and Networking Conference (WCNC), Nanjing, China, 29 March–1 April 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 1–6.
140. Khan, L.U.; Alsenwi, M.; Yaqoob, I.; Imran, M.; Han, Z.; Hong, C.S. Resource optimized Federated Learning-enabled cognitive internet of things for smart industries. *IEEE Access* **2020**, *8*, 168854–168864. [CrossRef]
141. Zhao, Y.; Li, L.; Liu, Y.; Fan, Y.; Lin, K.Y. Communication-efficient Federated Learning for digital twin systems of Industrial Internet of Things. *IFAC-PapersOnLine* **2022**, *55*, 433–438. [CrossRef]
142. Becker, S.; Styp-Rekowski, K.; Stoll, O.V.L.; Kao, O. Federated Learning for Autoencoder-based Condition Monitoring in the Industrial Internet of Things. In Proceedings of the 2022 IEEE International Conference on Big Data (Big Data), Osaka, Japan, 17–20 December 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 5424–5433.
143. Al-Abiad, M.S.; Hassan, M.Z.; Hossain, M.J. Energy efficient Federated Learning in integrated fog-cloud computing enabled Internet-of-Things networks. *arXiv* **2021**, arXiv:2107.03520.
144. Lee, J.; Ko, H. Energy and Distribution-Aware Cooperative Clustering Algorithm in Internet of Things (IoT)-Based Federated Learning. *IEEE Trans. Veh. Technol.* **2023**, *72*, 13799–13804. [CrossRef]
145. Wen, M.; Xie, R.; Lu, K.; Wang, L.; Zhang, K. Feddetect: A novel privacy-preserving Federated Learning framework for energy theft detection in smart grid. *IEEE Internet Things J.* **2021**, *9*, 6069–6080. [CrossRef]
146. Zhang, J.; Liu, Y.; Qin, X.; Xu, X. Energy-efficient Federated Learning framework for digital twin-enabled Industrial Internet of Things. In Proceedings of the 2021 IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Helsinki, Finland, 13–16 September 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 1160–1166.
147. Otoum, S.; Al Ridhawi, I.; Mouftah, H. A Federated Learning and Blockchain-Enabled Sustainable Energy Trade at the Edge: A Framework for Industry 4.0. *IEEE Internet Things J.* **2022**, *10*, 3018–3026. [CrossRef]
148. Badr, M.M.; Mahmoud, M.; Fang, Y.; Abdulaal, M.; Aljohani, A.J.; Alasmary, W.; Ibrahem, M.I. Privacy-preserving and communication-efficient energy prediction scheme based on Federated Learning for smart grids. *IEEE Internet Things J.* **2023**, *10*, 7719–7736. [CrossRef]
149. Hamdi, R.; Said, A.B.; Baccour, E.; Erbad, A.; Mohamed, A.; Hamdi, M.; Guizani, M. Optimal Resource Management for Hierarchical Federated Learning over HetNets with Wireless Energy Transfer. *IEEE Internet Things J.* **2023**, *10*, 16945–16958. [CrossRef]
150. Albelaihi, R.; Yu, L.; Craft, W.D.; Sun, X.; Wang, C.; Gazda, R. Green Federated Learning via energy-aware client selection. In Proceedings of the GLOBECOM 2022–2022 IEEE Global Communications Conference, Rio de Janeiro, Brazil, 4–8 December 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 13–18.
151. He, Y.; Luo, M.; Wu, B.; Sun, L.; Wu, Y.; Liu, Z.; Xiao, K. A Game Theory-Based Incentive Mechanism for Collaborative Security of Federated Learning in Energy Blockchain Environment. *IEEE Internet Things J.* **2023**, *10*, 21294–21308. [CrossRef]
152. Ahmed, U.; Srivastava, G.; Lin, J.C.W. Reliable customer analysis using Federated Learning and exploring deep-attention edge intelligence. *Future Gener. Comput. Syst.* **2022**, *127*, 70–79. [CrossRef]
153. Wang, H.; Xie, F.; Duan, Q.; Li, J. Federated Learning for Supply Chain Demand Forecasting. *Math. Probl. Eng.* **2022**, *2022*, 4109070. [CrossRef]
154. Xiao, B.; Xu, Q.; He, C.; Lin, J. Blockchain and Federated Learning Based Bidding Applications in Power Markets. *Procedia Comput. Sci.* **2022**, *202*, 21–26. [CrossRef]
155. Wang, Y.; Bennani, I.L.; Liu, X.; Sun, M.; Zhou, Y. Electricity consumer characteristics identification: A federated learning approach. *IEEE Trans. Smart Grid* **2021**, *12*, 3637–3647. [CrossRef]
156. Lin, J.; Ma, J.; Zhu, J. Privacy-preserving household characteristic identification with Federated Learning method. *IEEE Trans. Smart Grid* **2021**, *13*, 1088–1099. [CrossRef]
157. Wang, Y.; Jia, M.; Gao, N.; Von Krannichfeldt, L.; Sun, M.; Hug, G. Federated clustering for electricity consumption pattern extraction. *IEEE Trans. Smart Grid* **2022**, *13*, 2425–2439. [CrossRef]
158. Husnoo, M.A.; Anwar, A.; Hosseinzadeh, N.; Islam, S.N.; Mahmood, A.N.; Doss, R. FedREP: Towards horizontal federated load forecasting for retail energy providers. In Proceedings of the 2022 IEEE PES 14th Asia-Pacific Power and Energy Engineering Conference (APPEEC), Melbourne, Australia, 20–23 November 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 1–6.
159. Liu, C.; Guo, S.; Guo, S.; Yan, Y.; Qiu, X.; Zhang, S. Ltsm: Lightweight and trusted sharing mechanism of iot data in smart city. *IEEE Internet Things J.* **2021**, *9*, 5080–5093. [CrossRef]
160. Guo, S.; Xiang, B.; Chen, L.; Yang, H.; Yu, D. Multi-level Federated Learning Mechanism with Reinforcement Learning Optimizing in Smart City. In Proceedings of the International Conference on Artificial Intelligence and Security, Qinghai, China, 15–20 July 2022; Springer: Berlin/Heidelberg, Germany, 2022; pp. 441–454.
161. Kuang, Z.; Chen, C. Research on smart city data encryption and communication efficiency improvement under Federated Learning framework. *Egypt. Inform. J.* **2023**, *24*, 217–227. [CrossRef]

162. Zhang, S.; Duan, J.; Liu, H. Online task assignment for Federated Learning in smart city. In Proceedings of the 2nd International Conference on Internet of Things and Smart City (IoTSC 2022), Xiamen, China, 18–20 February 2022; SPIE: Bellingham, WA, USA, 2022; Volume 12249, pp. 509–517.
163. Dey, S.; Pal, S. Federated Learning-based air quality prediction for smart cities using BGRU model. In Proceedings of the 28th Annual International Conference on Mobile Computing And Networking, Sydney, NSW, Australia, 17–21 October 2022; pp. 871–873.
164. Siddiqui, S.; Khan, A.A.; Dev, K.; Dey, I. Integrating Federated Learning with IoMT for managing obesity in smart city. In Proceedings of the 1st Workshop on Artificial Intelligence and Blockchain Technologies for Smart Cities with 6G, New Orleans, LA, USA, 25–29 October 2021; pp. 7–12.
165. Qolomany, B.; Ahmad, K.; Al-Fuqaha, A.; Qadir, J. Particle swarm optimized Federated Learning for industrial IoT and smart city services. In Proceedings of the GLOBECOM 2020–2020 IEEE Global Communications Conference, Taipei, Taiwan, 7–11 December 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 1–6.
166. Mitra, A.; Ngoko, Y.; Trystram, D. Impact of Federated Learning on smart buildings. In Proceedings of the 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), Coimbatore, India, 25–27 March 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 93–99.
167. Guo, S.; Xiang, B.; Xia, X.; Yan, Z.; Li, Y. Blockchain and Federated Learning Based Data Security Sharing Mechanism over Smart City. 2020. Available online: https://www.researchgate.net/publication/346852381_Blockchain_and_Federated_Learning_Based_Data_Security_Sharing_Mechanism_Over_Smart_City (accessed on 1 October 2024).
168. Li, Y.; Yu, H.; Zeng, Y.; Pan, Q. HFSA: A Semi-asynchronous Hierarchical Federated Recommendation System in Smart City. *IEEE Internet Things J.* **2023**, *10*, 18808–18820. [CrossRef]
169. Zhong, X. Intelligent Chinese Classroom Teaching Mode Based on the Internet of Things and Federated Learning-Joint Optimization of Teaching Mode. *KeAi Int. J. Intell. Netw.* **2023**. [CrossRef]
170. Sun, H.; Li, S.; Yu, F.R.; Qi, Q.; Wang, J.; Liao, J. Toward communication-efficient Federated Learning in the Internet of Things with edge computing. *IEEE Internet Things J.* **2020**, *7*, 11053–11067. [CrossRef]
171. Cao, H.; Feng, W.; He, J.; Liu, S. Decentralized Federated Learning for Secure Space-Terrestrial Communication With Intelligent Reflecting Surface. *IEEE Wirel. Commun. Lett.* **2023**, *12*, 2083–2087. [CrossRef]
172. Lu, Y.; Huang, X.; Zhang, K.; Maharjan, S.; Zhang, Y. Communication-efficient Federated Learning and permissioned blockchain for digital twin edge networks. *IEEE Internet Things J.* **2020**, *8*, 2276–2288. [CrossRef]
173. Chen, Z.; Cui, H.; Wu, E.; Yu, X. Computation and communication efficient adaptive federated optimization of Federated Learning for Internet of Things. *Electronics* **2023**, *12*, 3451. [CrossRef]
174. Lim, W.Y.B.; Garg, S.; Xiong, Z.; Zhang, Y.; Niyato, D.; Leung, C.; Miao, C. UAV-assisted communication efficient Federated Learning in the era of the artificial intelligence of things. *IEEE Netw.* **2021**, *35*, 188–195. [CrossRef]
175. Ji, Y.; Chen, L. FedQNN: A Computation–Communication-Efficient Federated Learning Framework for IoT With Low-Bitwidth Neural Network Quantization. *IEEE Internet Things J.* **2022**, *10*, 2494–2507. [CrossRef]
176. Yao, J.; Ansari, N. Secure Federated Learning by power control for internet of drones. *IEEE Trans. Cogn. Commun. Netw.* **2021**, *7*, 1021–1031. [CrossRef]
177. Zhang, Z.; Ma, S.; Yang, Z.; Xiong, Z.; Kang, J.; Wu, Y.; Zhang, K.; Niyato, D. Robust Semisupervised Federated Learning for Images Automatic Recognition in Internet of Drones. *IEEE Internet Things J.* **2022**, *10*, 5733–5746. [CrossRef]
178. Islam, A.; Shin, S.Y. Fedrc: A Federated Learning-based roadside computing paradigm through the facilitation of internet of drones. In Proceedings of the 2021 International Conference on Information and Communication Technology Convergence (ICTC), Korea, Republic of Korea, 20–22 October 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 648–653.
179. Alsamhi, S.H.; Almalki, F.A.; Afghah, F.; Hawbani, A.; Shvetsov, A.V.; Lee, B.; Song, H. Drones' edge intelligence over smart environments in B5G: Blockchain and Federated Learning synergy. *IEEE Trans. Green Commun. Netw.* **2021**, *6*, 295–312. [CrossRef]
180. Al-Abiad, M.S.; Hossain, M.J. Coordinated Scheduling and Decentralized Federated Learning Using Conflict Clustering Graphs in Fog-Assisted IoD Networks. *IEEE Trans. Veh. Technol.* **2022**, *72*, 3455–3472. [CrossRef]
181. Chhikara, P.; Tekchandani, R.; Kumar, N.; Tanwar, S. Federated Learning-based aerial image segmentation for collision-free movement and landing. In Proceedings of the 4th ACM MobiCom Workshop on Drone Assisted Wireless Communications for 5G and Beyond, Virtual, 29 October 2021; pp. 13–18.
182. Pokhrel, S.R.; Mandjes, M. Internet of Drones: Improving Multipath TCP over WiFi with Federated Multi-Armed Bandits for Limitless Connectivity. *Drones* **2022**, *7*, 30. [CrossRef]
183. Li, H.; Zhang, H.; Liao, D.; Zhu, X.; Dai, Y.; Verma, S. A data sharing method for internet of drones based on federated learning. In Proceedings of the 5th International ACM Mobicom Workshop on Drone Assisted Wireless Communications for 5G and Beyond, Sydney, NSW, Australia, 17 October 2022; pp. 91–96.
184. Yao, J.; Sun, X. Energy-Efficient Federated Learning in Internet of Drones Networks. In Proceedings of the 2023 IEEE 24th International Conference on High Performance Switching and Routing (HPSR), Albuquerque, NM, USA, 5–7 June 2023; IEEE: Piscataway, NJ, USA, 2023; pp. 185–190.
185. Xu, Z.; Guo, Y.; Chakraborty, C.; Hua, Q.; Chen, S.; Yu, K. A simple Federated Learning-based scheme for security enhancement over Internet of Medical Things. *IEEE J. Biomed. Health Inform.* **2022**, *27*, 652–663. [CrossRef] [PubMed]

186. Fan, J.; Wang, X.; Guo, Y.; Hu, X.; Hu, B. Federated Learning driven secure internet of medical things. *IEEE Wirel. Commun.* **2022**, *29*, 68–75. [CrossRef]
187. Wang, X.; Hu, J.; Lin, H.; Liu, W.; Moon, H.; Piran, M.J. Federated Learning-empowered disease diagnosis mechanism in the internet of medical things: From the privacy-preservation perspective. *IEEE Trans. Ind. Inform.* **2022**, *19*, 7905–7913. [CrossRef]
188. Kwon, D.; Jeon, J.; Park, S.; Kim, J.; Cho, S. Multiagent DDPG-based deep learning for smart ocean federated learning IoT networks. *IEEE Internet Things J.* **2020**, *7*, 9895–9903.[CrossRef]
189. Fadlullah, Z.M.; Kato, N. On smart IoT remote sensing over integrated terrestrial-aerial-space networks: An asynchronous Federated Learning approach. *IEEE Netw.* **2021**, *35*, 129–135. [CrossRef]
190. Islam, A.; Al Amin, A.; Shin, S.Y. FBI: A Federated Learning-based blockchain-embedded data accumulation scheme using drones for Internet of Things. *IEEE Wirel. Commun. Lett.* **2022**, *11*, 972–976. [CrossRef]

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